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Opie et al.

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(54) **MEDICAL DEVICE FOR SENSING AND OR STIMULATING TISSUE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 507 days.

This patent is subject to a terminal disclaimer.

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(65) **Prior Publication Data**

US 2021/0393948 A1 Dec. 23, 2021

Related U.S. Application Data

(63) Continuation of application No. 16/539,357, filed on Aug. 13, 2019, now Pat. No. 11,141,584, which is a (Continued)

(30) **Foreign Application Priority Data**

Oct. 20, 2015 (AU) 2015904302
Dec. 4, 2015 (AU) 2015905045

(51) **Int. Cl.**
A61N 1/05 (2006.01)
A61B 5/293 (2021.01)

(Continued)

(52) **U.S. Cl.**
CPC **A61N 1/0531** (2013.01); **A61B 5/293** (2021.01); **A61F 2/91** (2013.01); **A61F 2/92** (2013.01);

(Continued)

(58) **Field of Classification Search**
CPC A61N 1/0531
(Continued)

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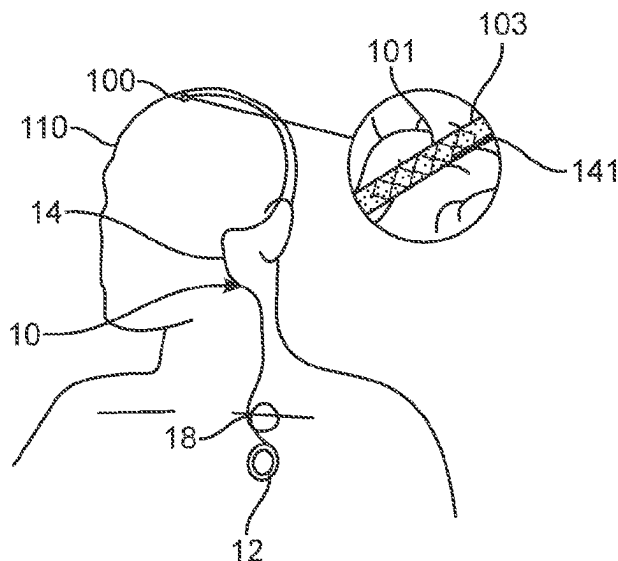
Primary Examiner — Nicole F Johnson

(74) *Attorney, Agent, or Firm* — Levine Bagade Han LLP

(57) **ABSTRACT**

Devices, methods and systems for transmitting signals through a device located in a blood vessel of an animal, for stimulating and/or sensing activity of media proximal to the device, wherein the media includes tissue and/or fluid.

22 Claims, 30 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/957,574, filed on Apr. 19, 2018, now Pat. No. 10,485,968, which is a continuation of application No. PCT/US2016/057768, filed on Oct. 19, 2016.

- (60) Provisional application No. 62/379,625, filed on Aug. 25, 2016.

- (51) **Int. Cl.**
A61F 2/91 (2013.01)
A61F 2/92 (2013.01)
A61B 18/00 (2006.01)

- (52) **U.S. Cl.**
 CPC *A61B 2018/00214* (2013.01); *A61B 2018/00267* (2013.01); *A61F 2250/0001* (2013.01); *A61F 2250/0002* (2013.01); *A61N 1/0534* (2013.01); *A61N 1/0539* (2013.01)

- (58) **Field of Classification Search**
 USPC 607/116
 See application file for complete search history.

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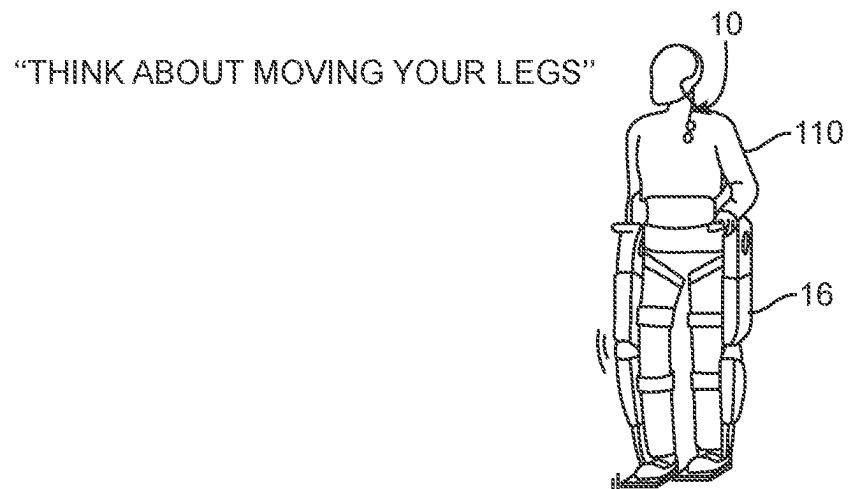


FIG. 1

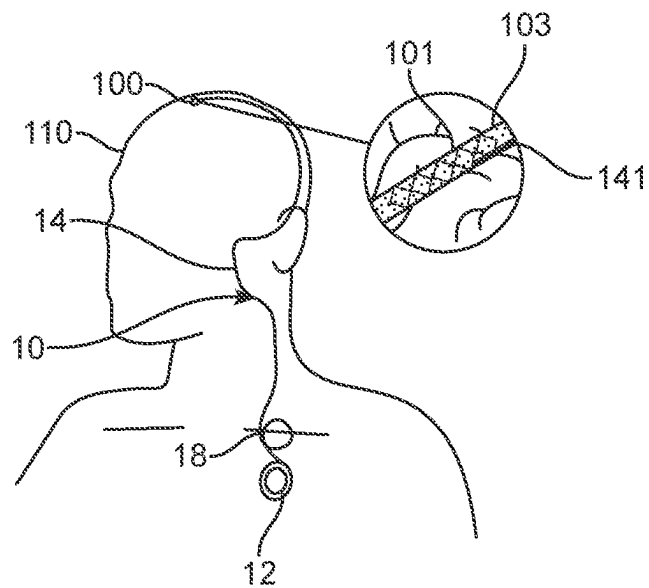


FIG. 2A

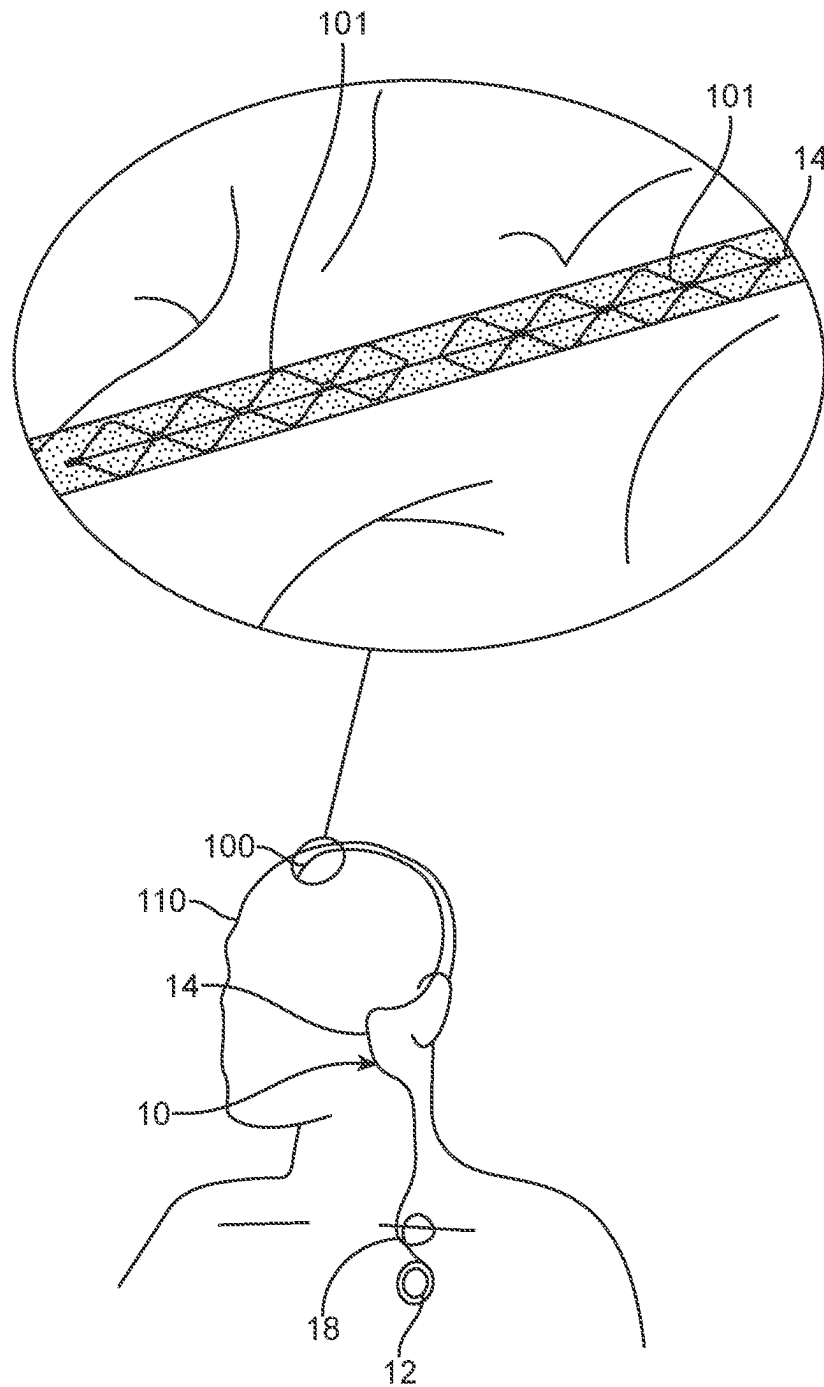


FIG. 2B

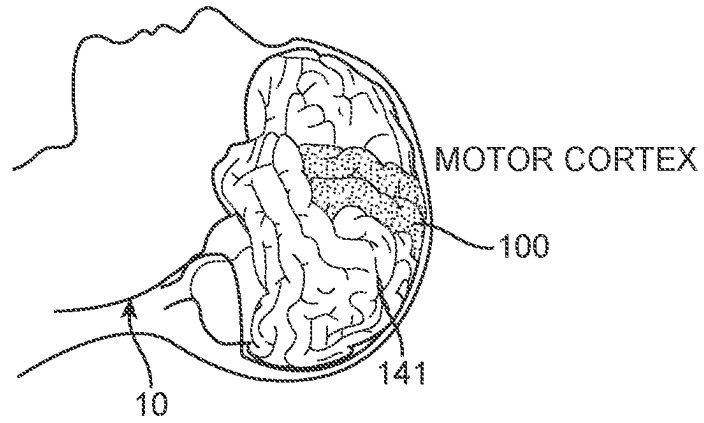


FIG. 3

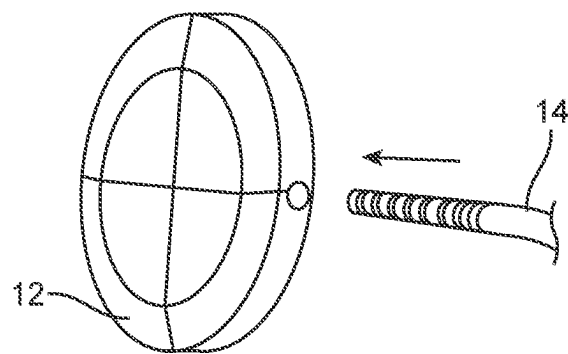


FIG. 4

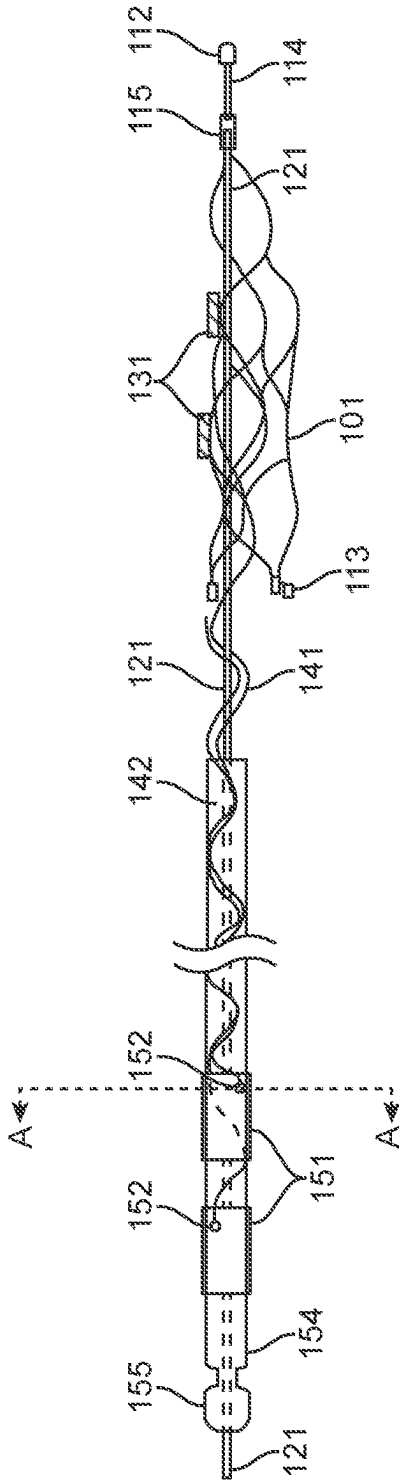


FIG. 5A

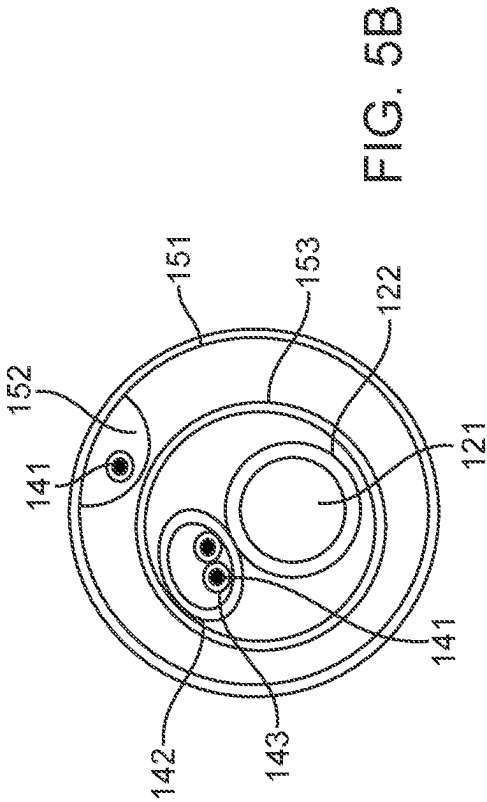


FIG. 5B

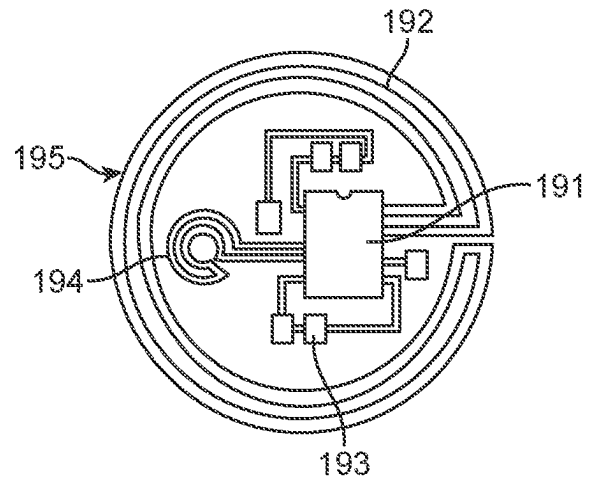


FIG. 5C

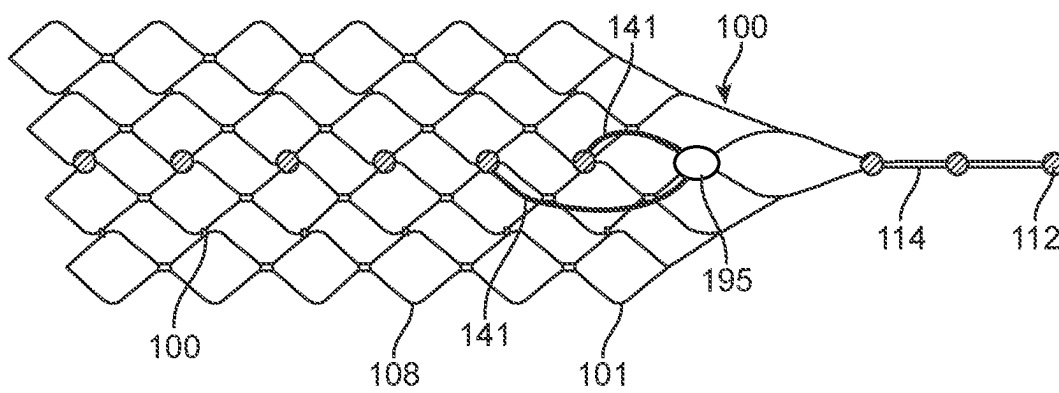


FIG. 5D

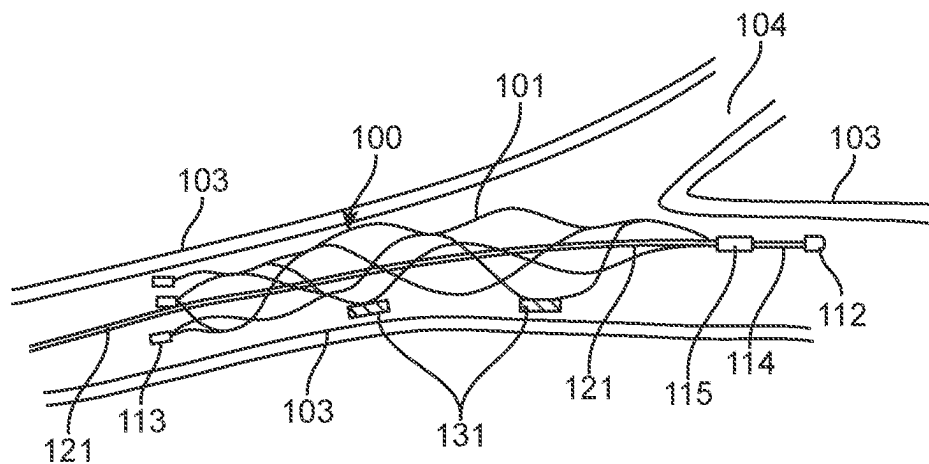


FIG. 6

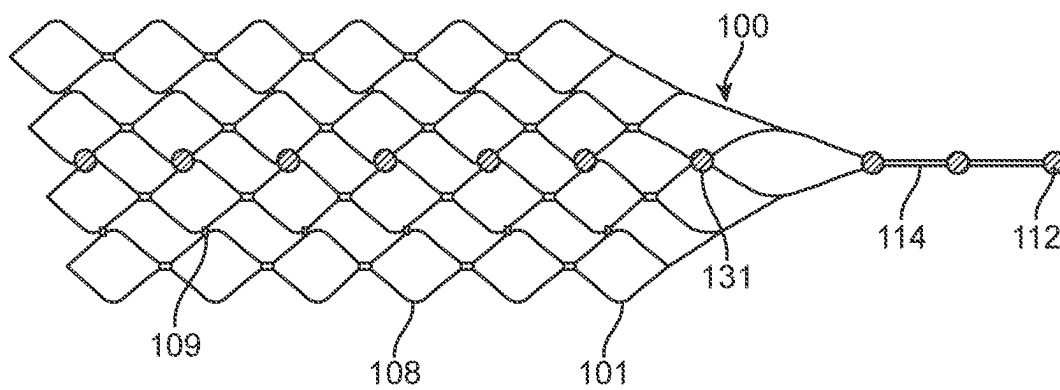


FIG. 7A

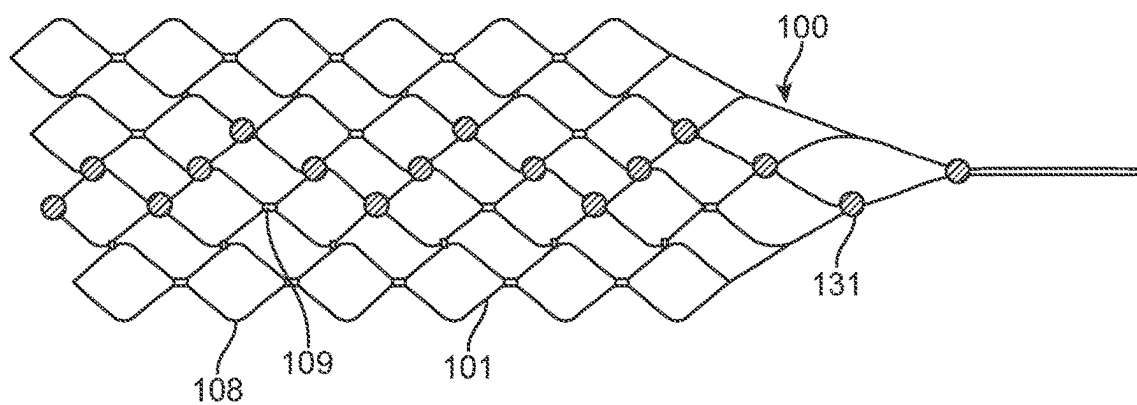


FIG. 7B

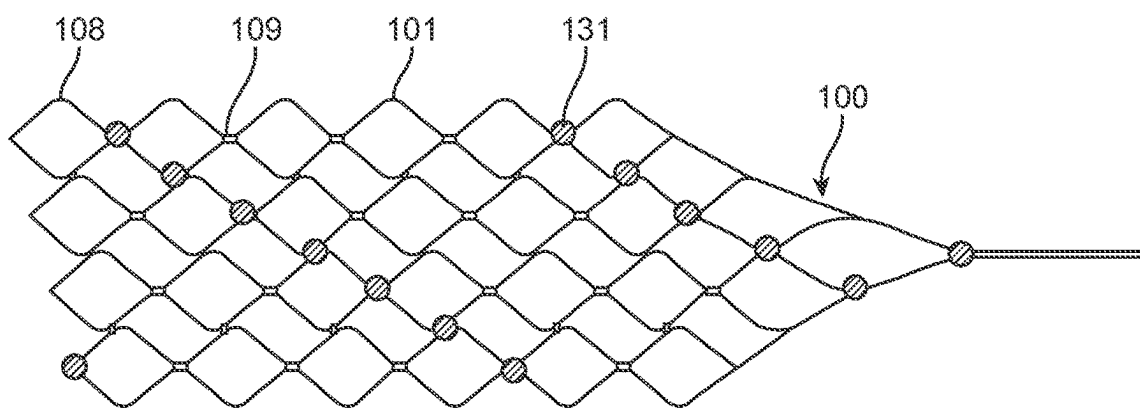


FIG. 7C

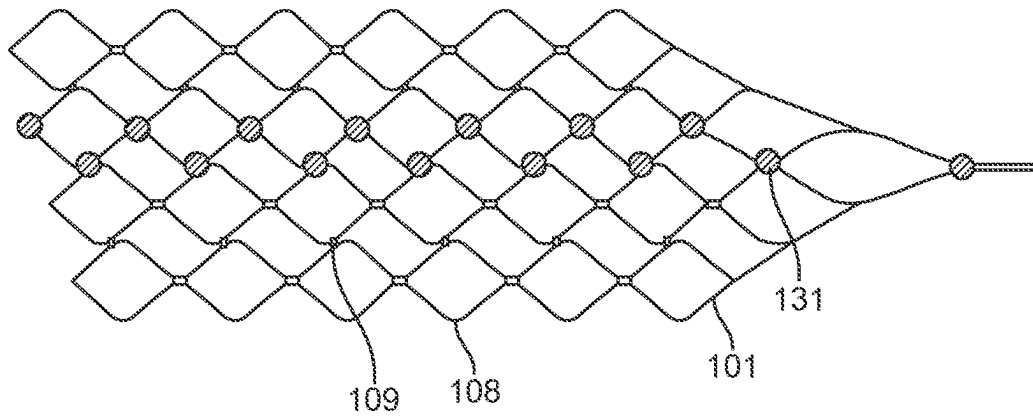


FIG. 7D

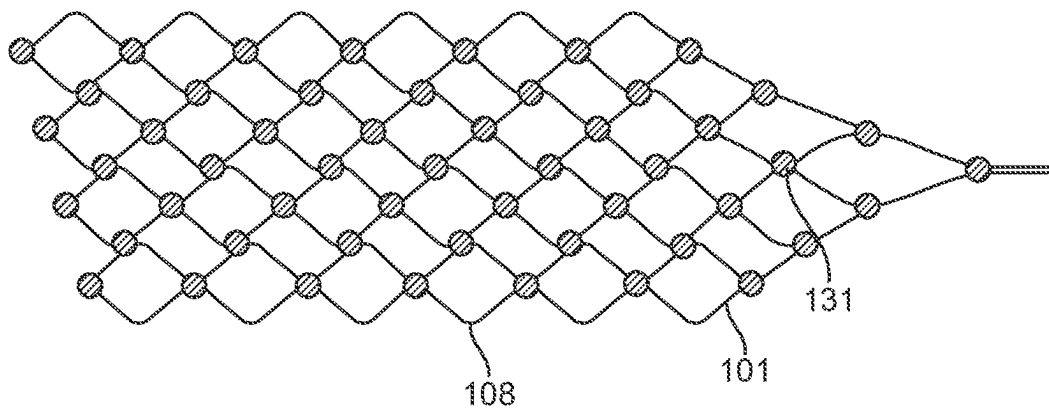


FIG. 7E

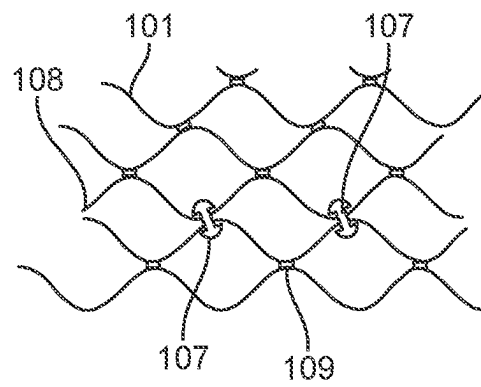


FIG. 8A

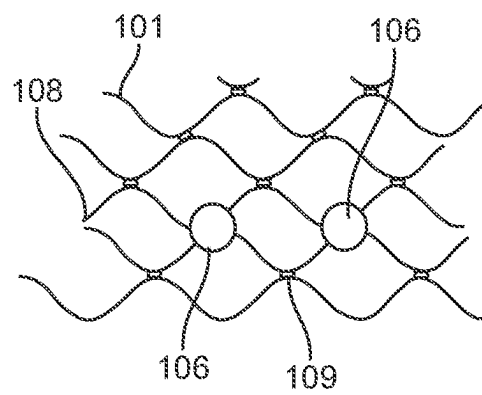


FIG. 8B

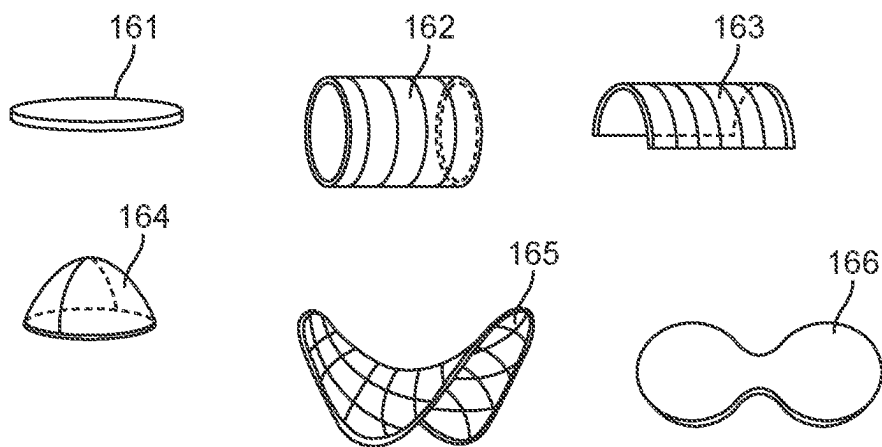


FIG. 9

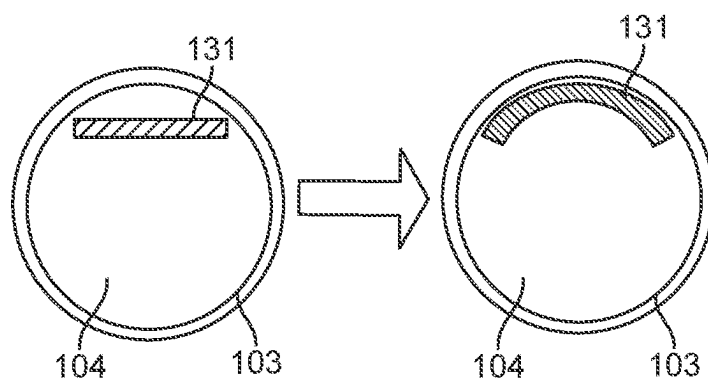


FIG. 10

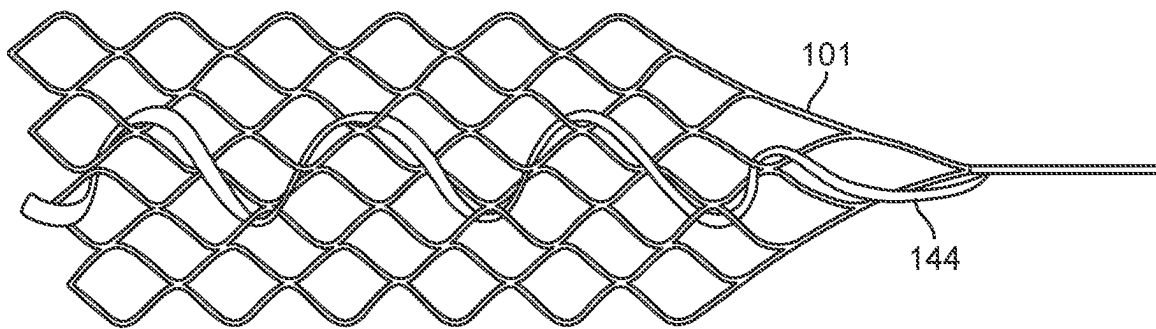


FIG. 11

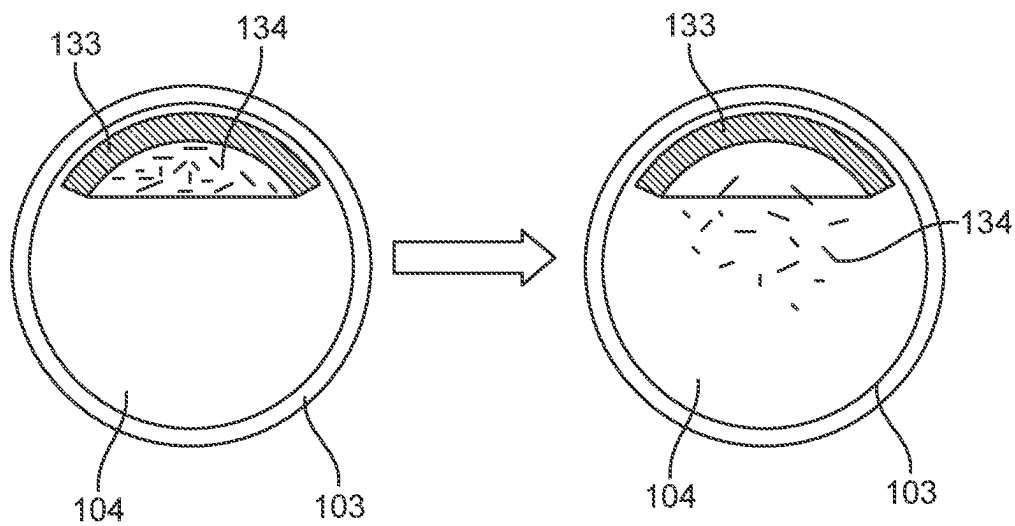


FIG. 12

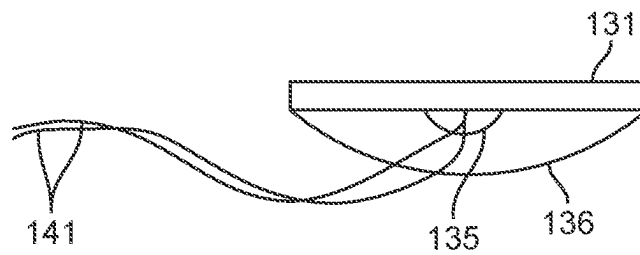


FIG. 13a

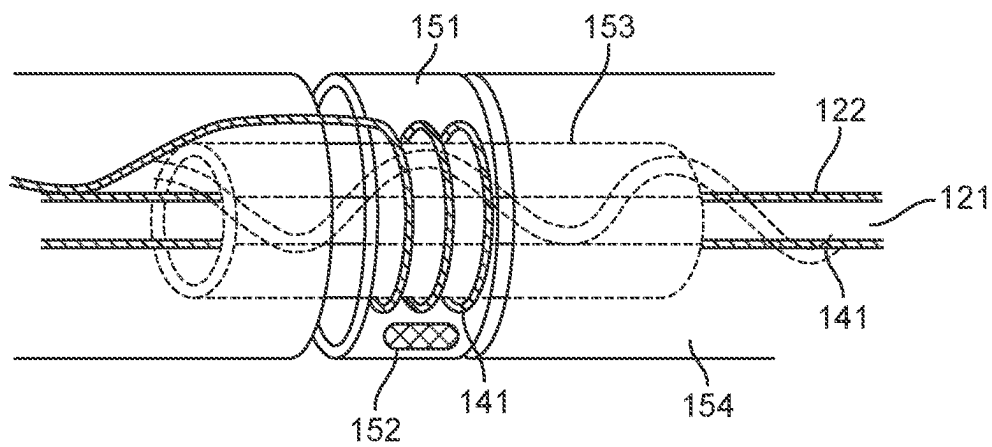


FIG. 13b

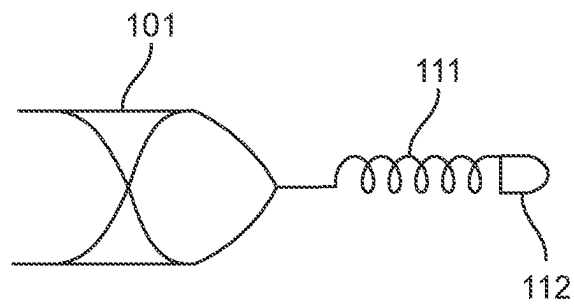


FIG. 14

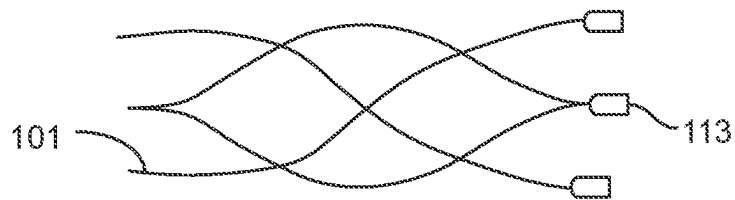


FIG. 15

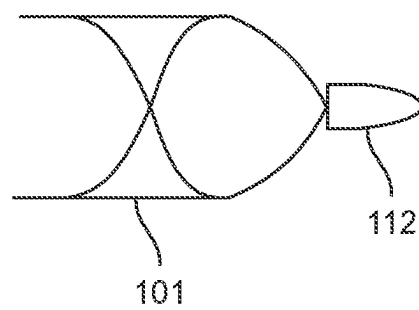


FIG. 16

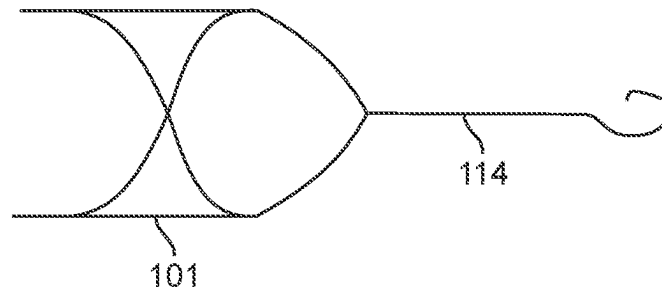


FIG. 17

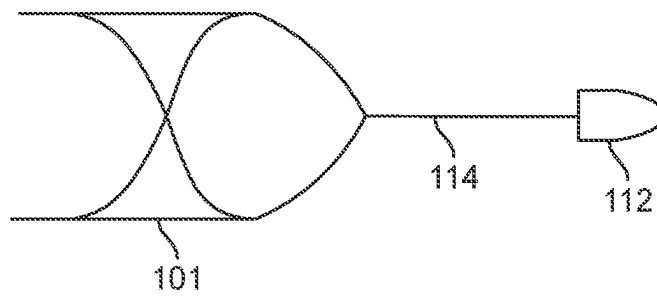


FIG. 18

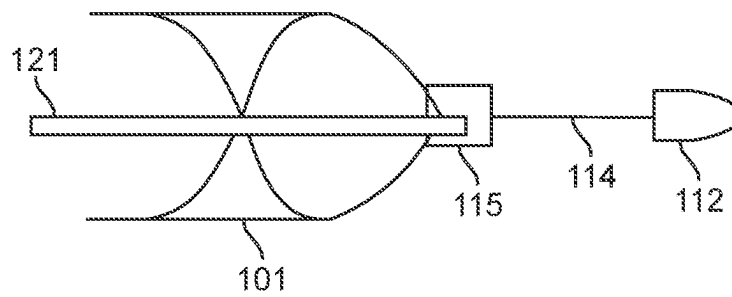


FIG. 19

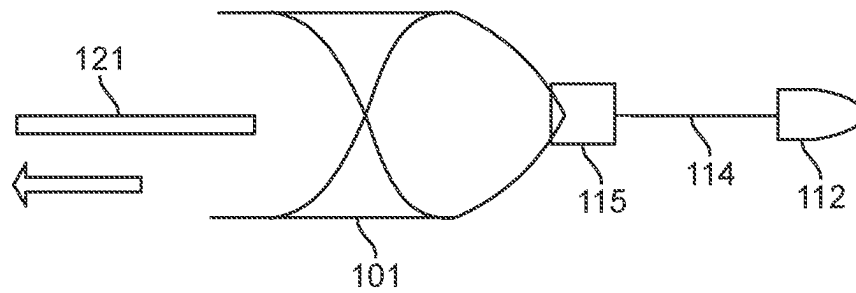


FIG. 20

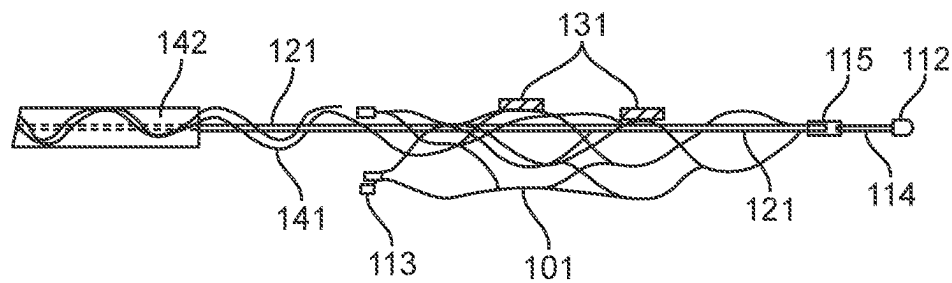


FIG. 21a

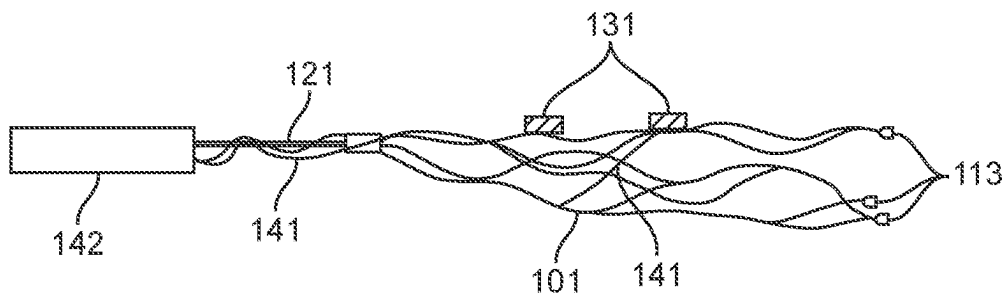


FIG. 21b

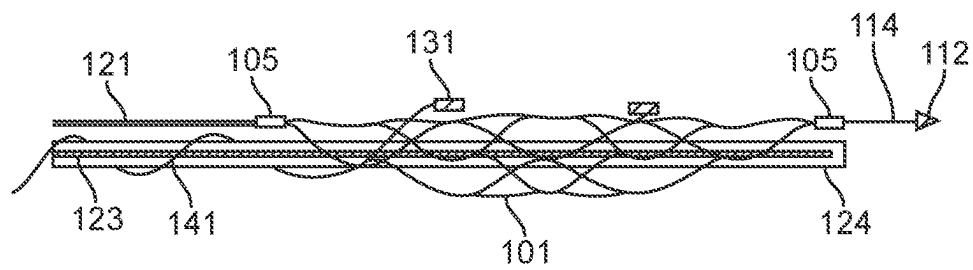


FIG. 21c

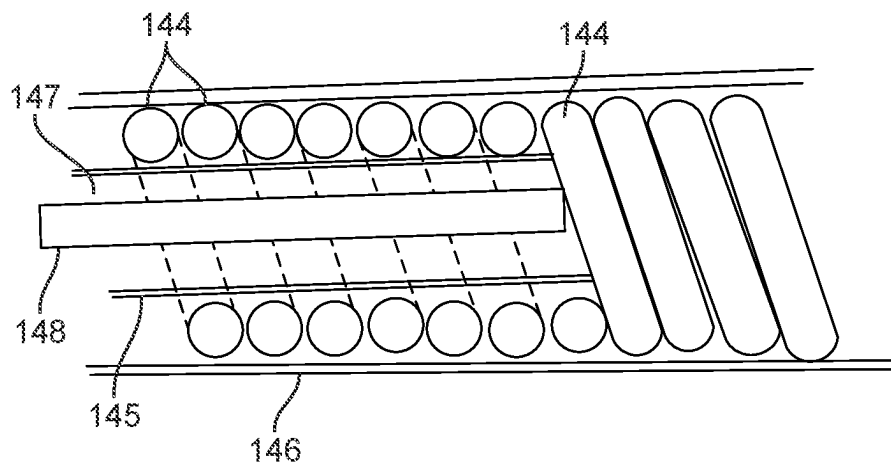


FIG. 21d

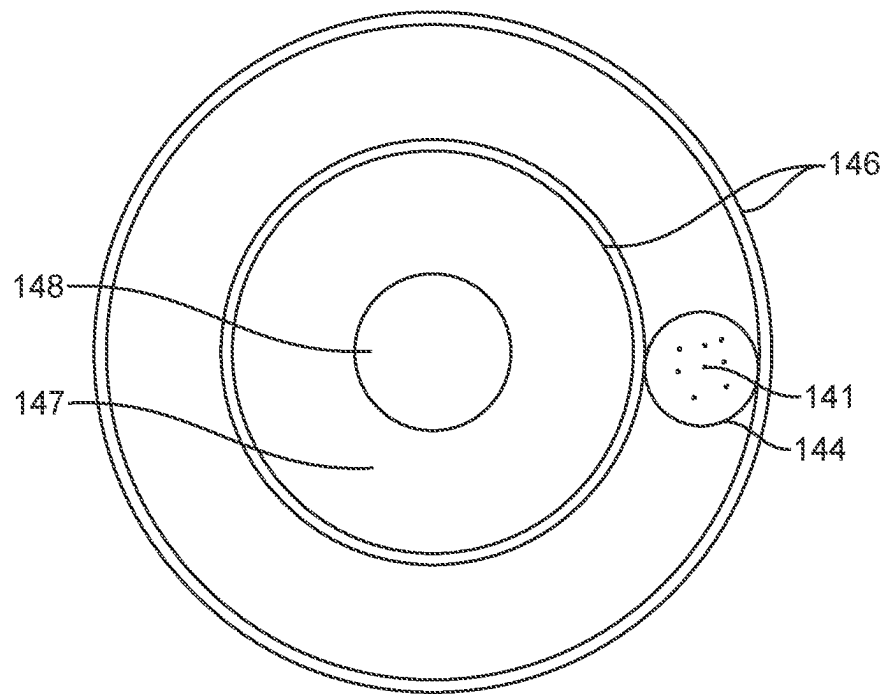


FIG. 21e

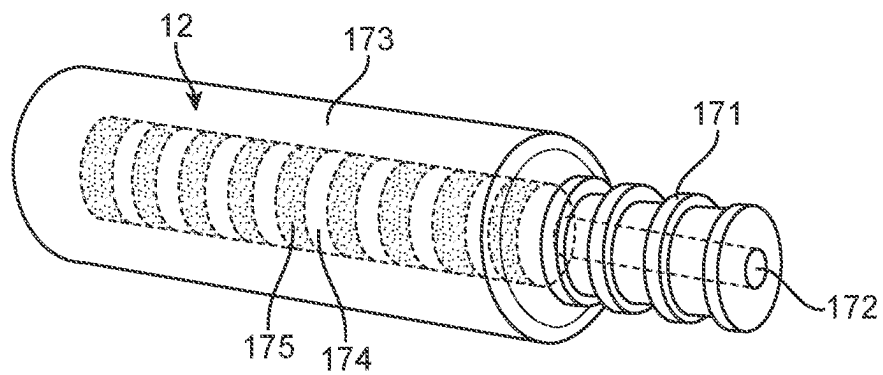


FIG. 22

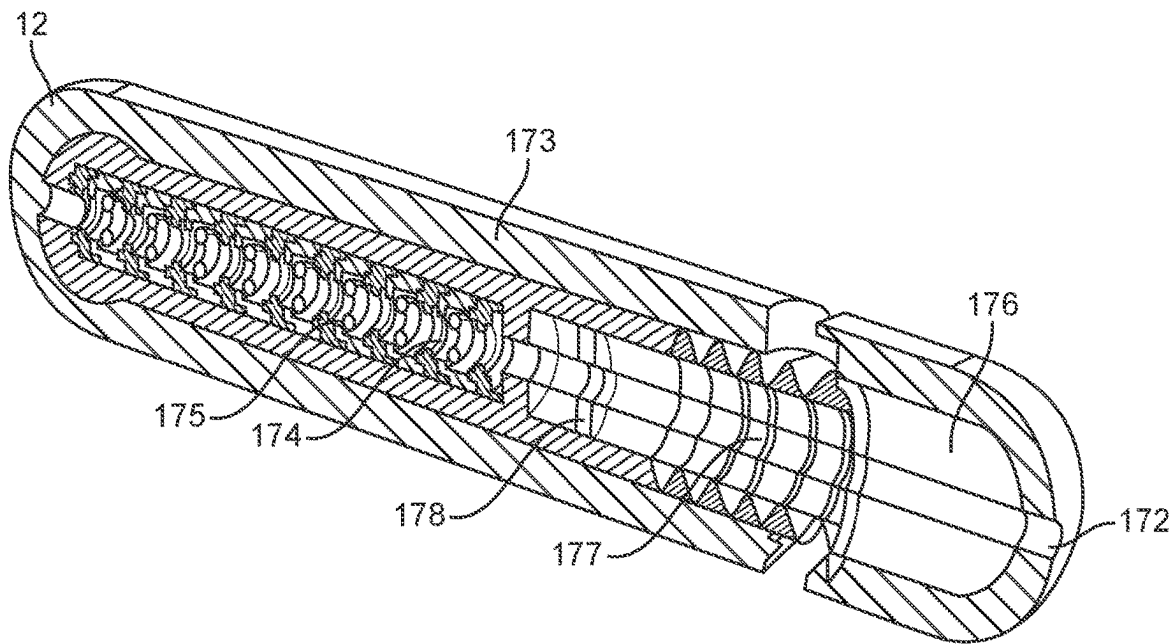


FIG. 23

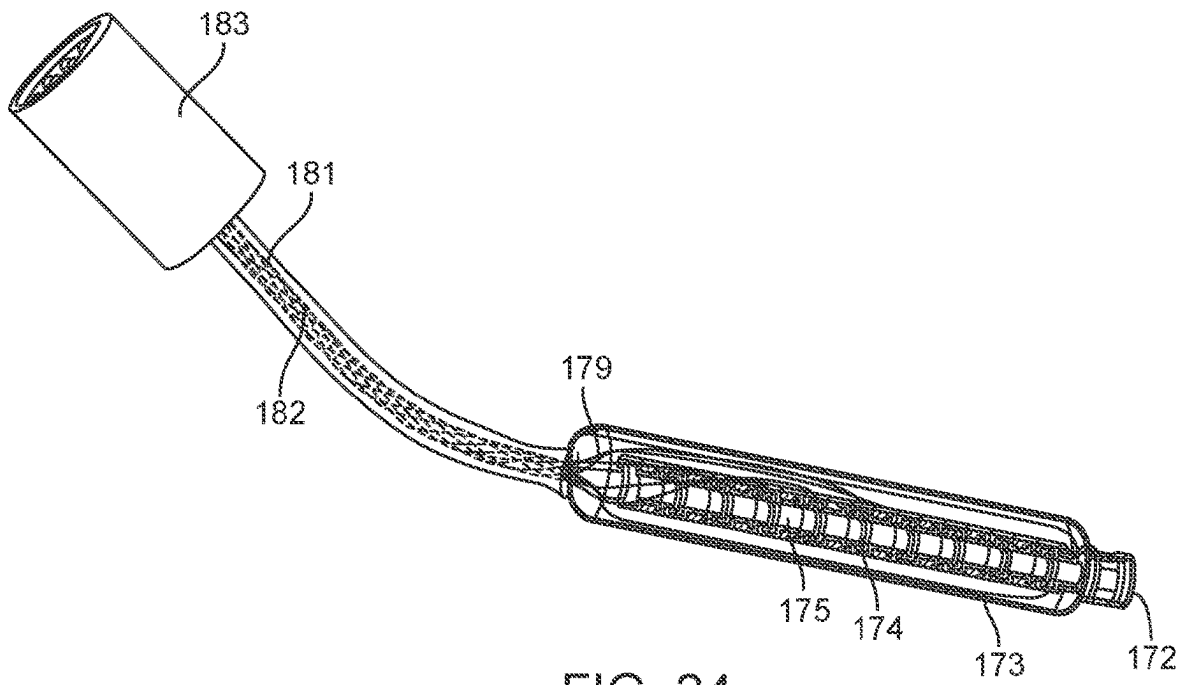


FIG. 24

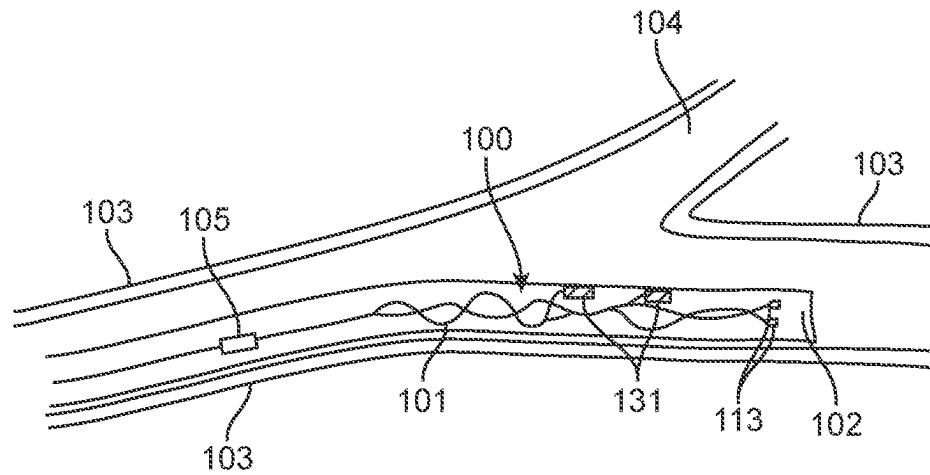


FIG. 25

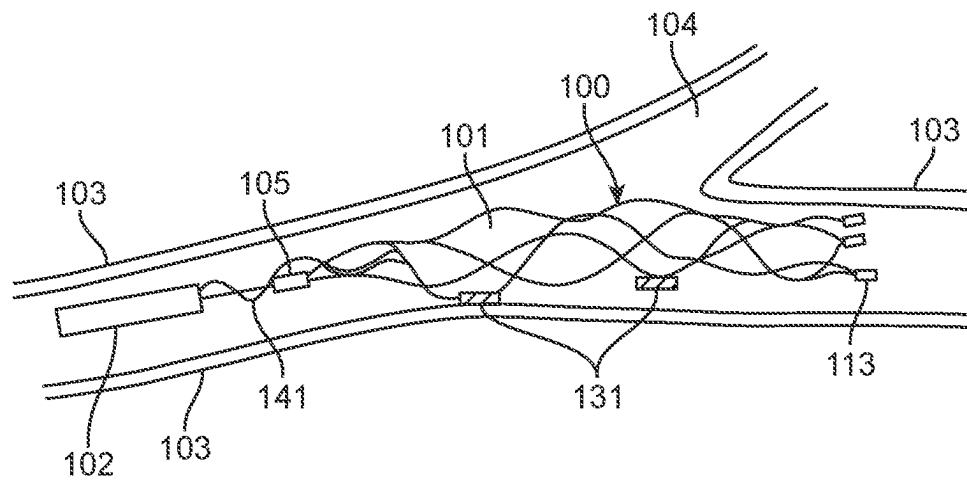


FIG. 26

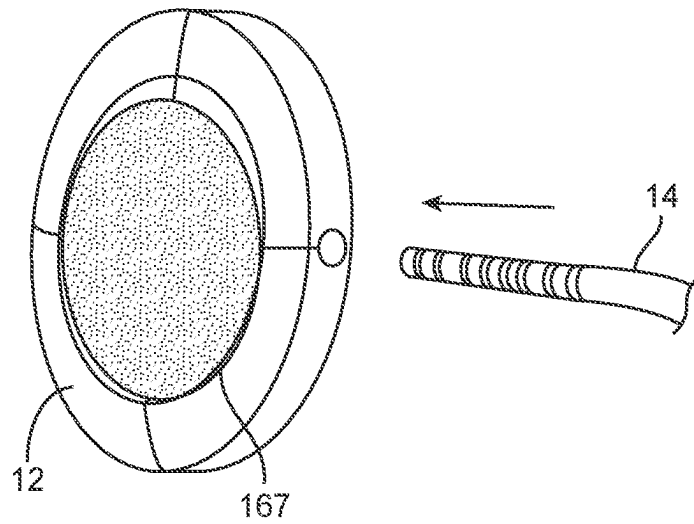


FIG. 27

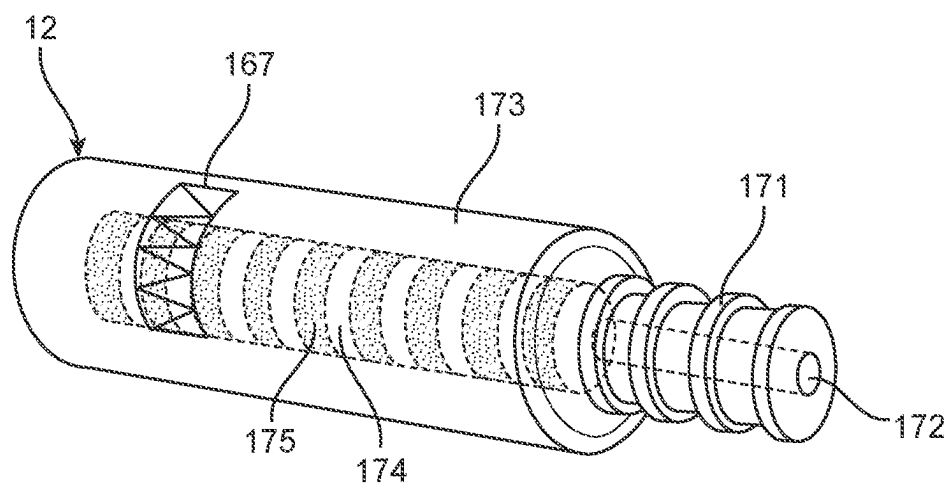


FIG. 28

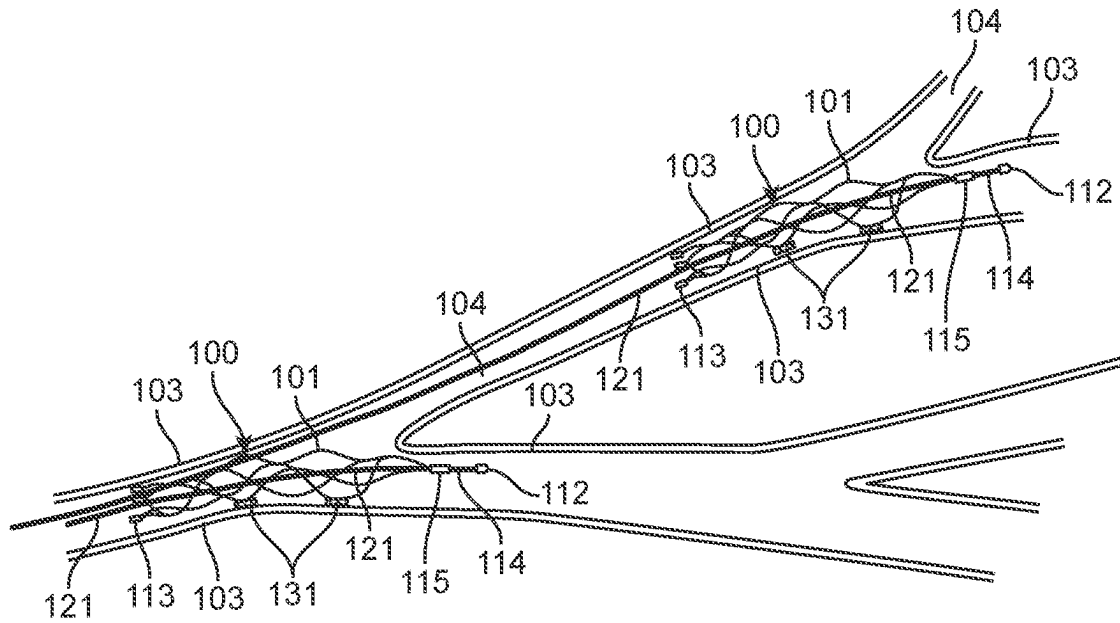


FIG. 29

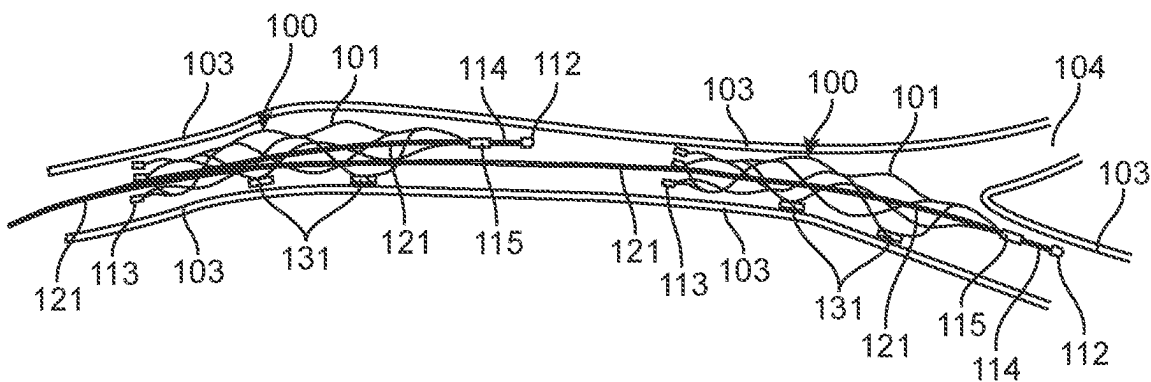


FIG. 30

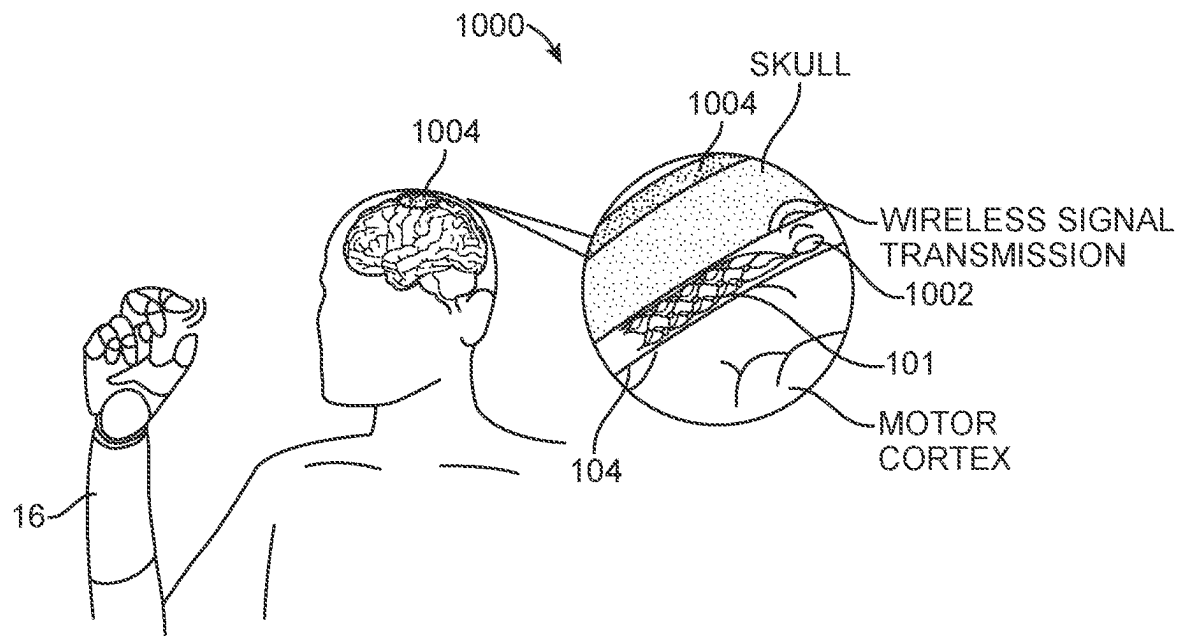


FIG. 31

Think about using your legs

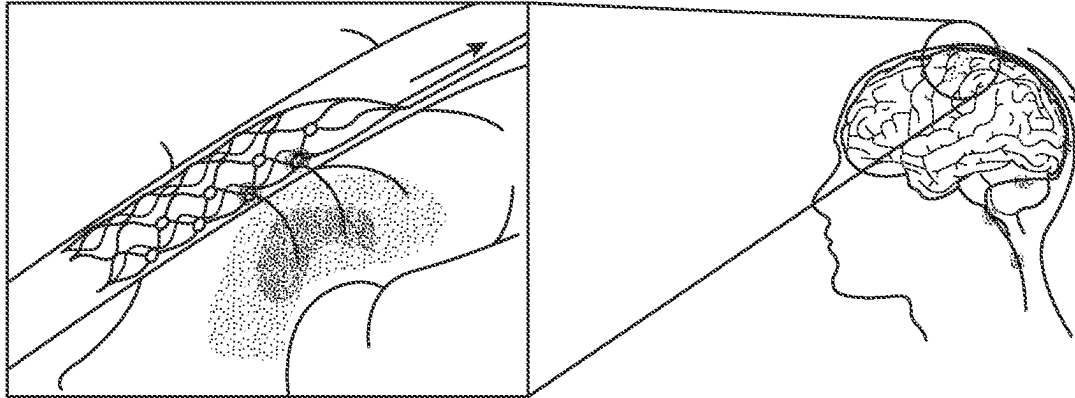


FIG. 32

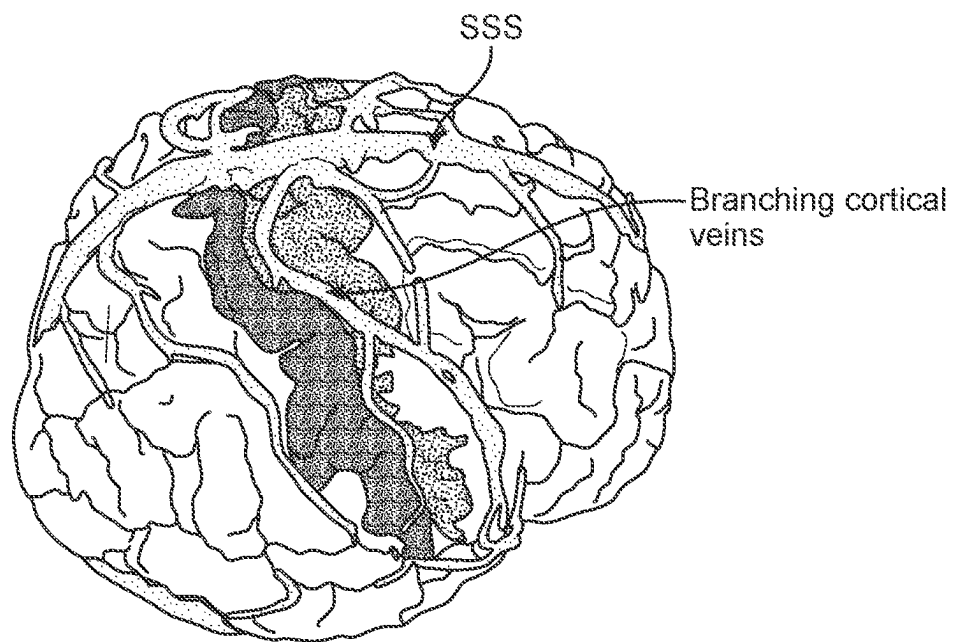


FIG. 33

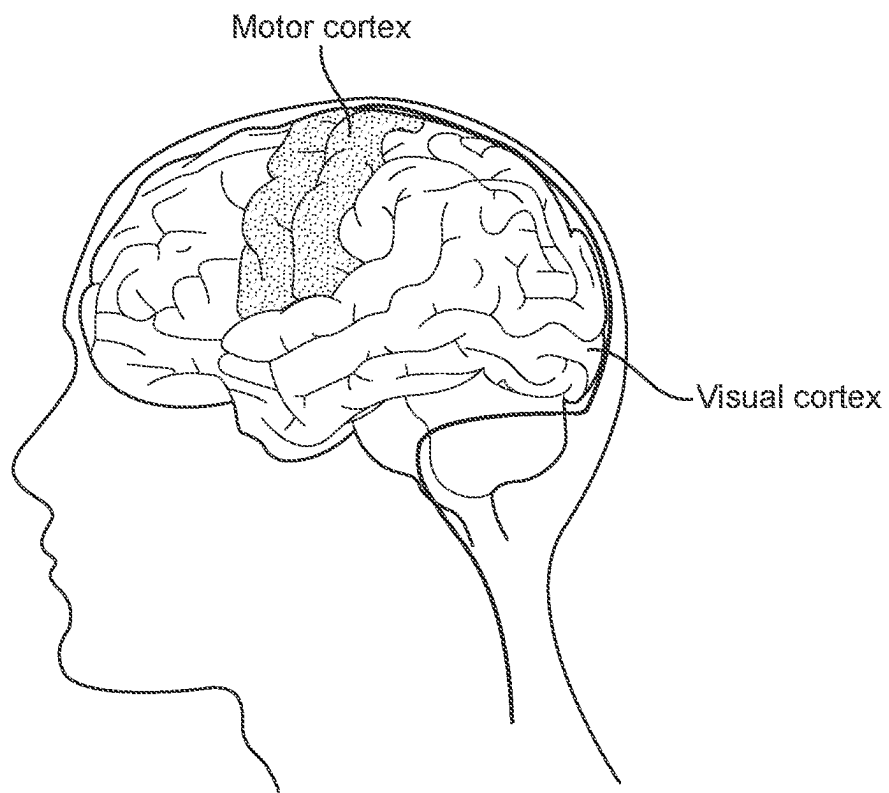


FIG. 34

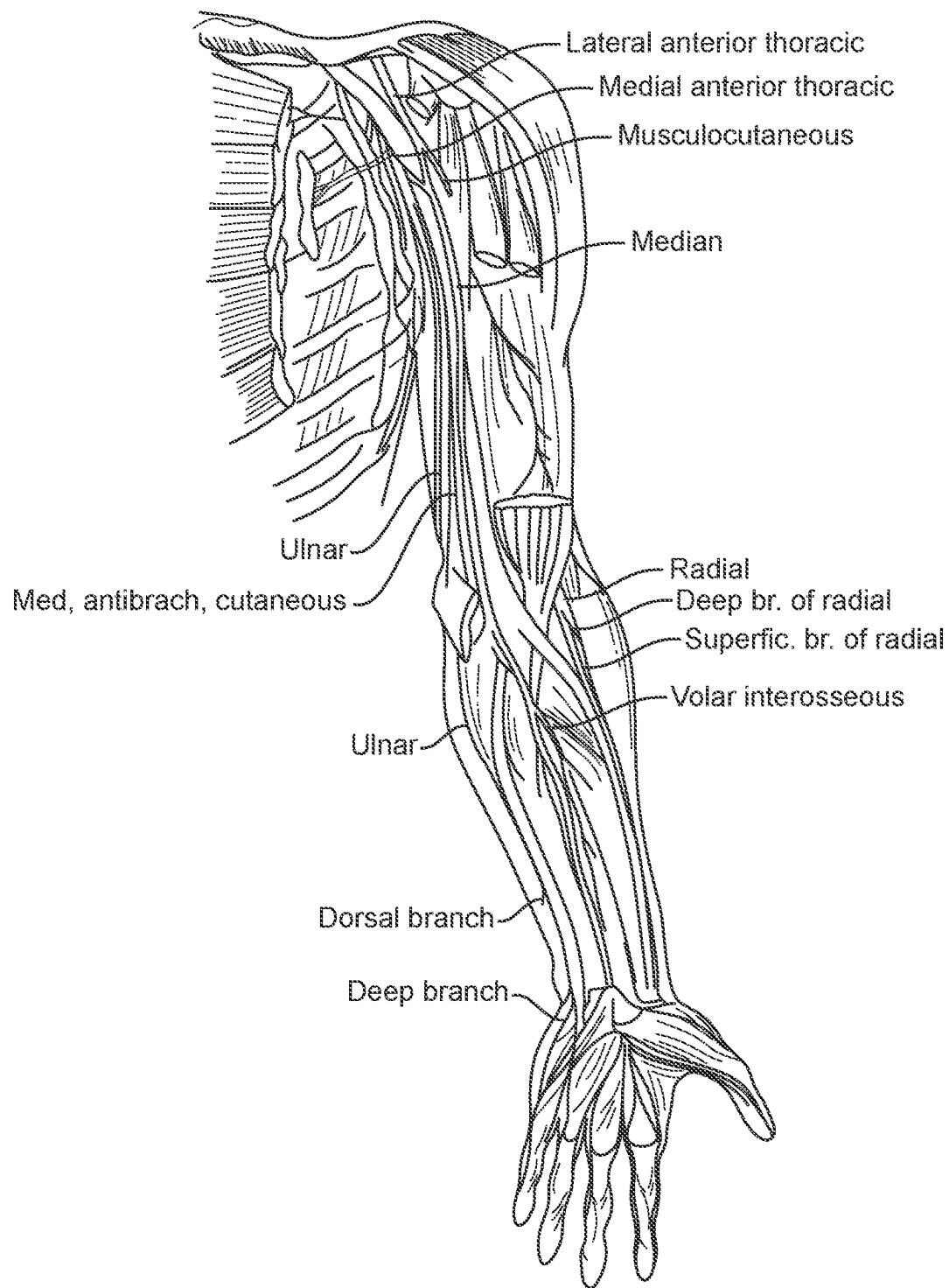


FIG. 35

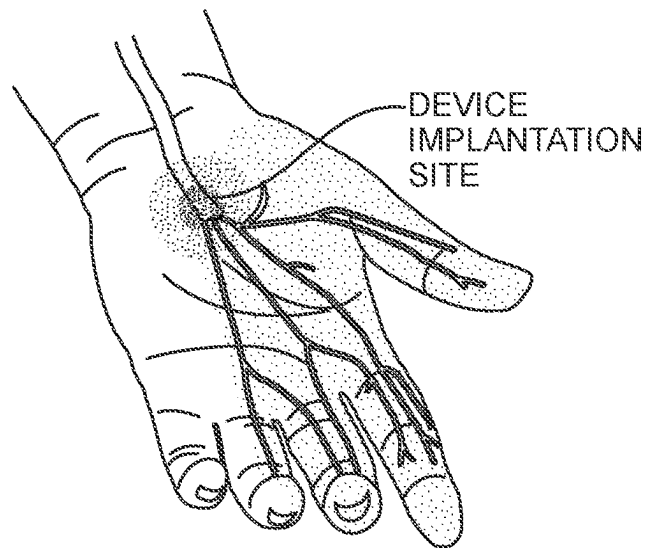


FIG. 36

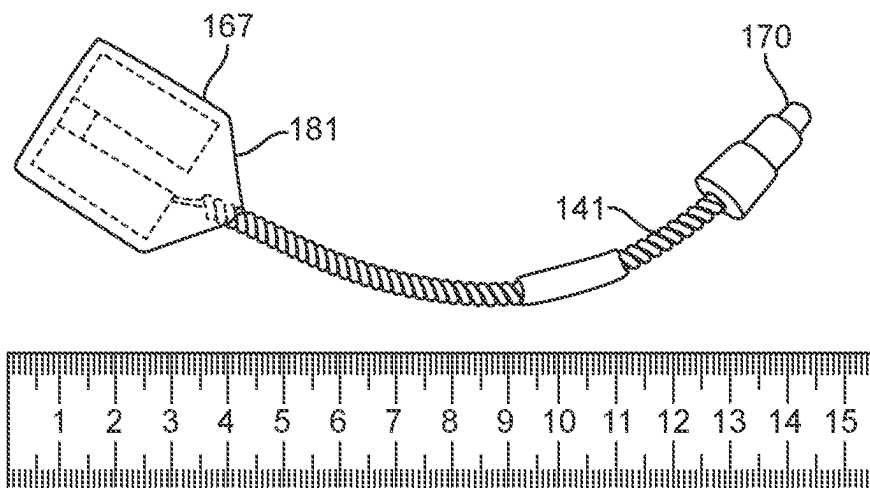


FIG. 37

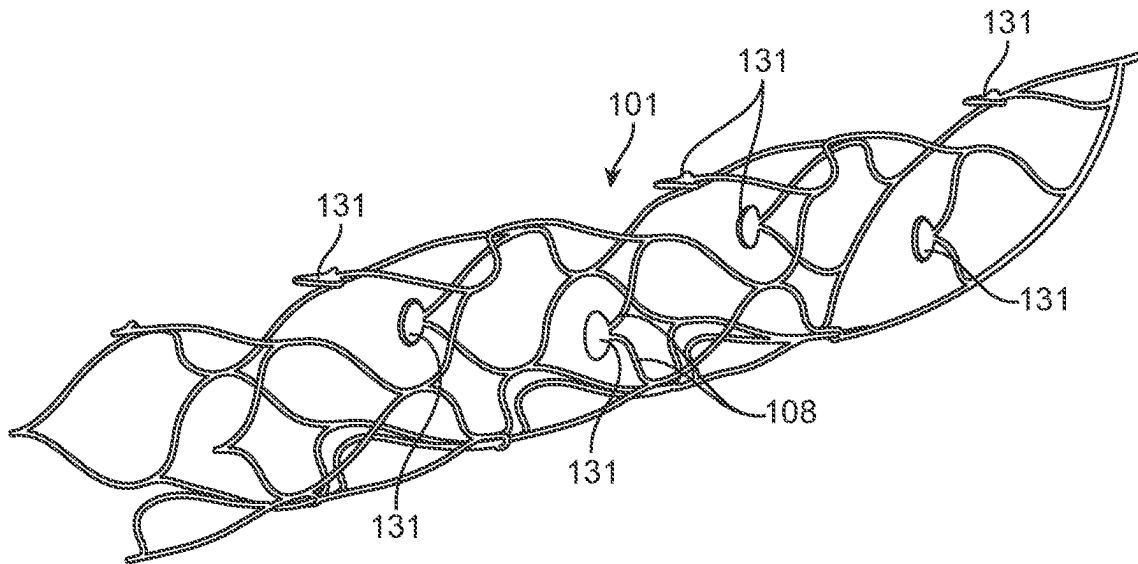


FIG. 38A

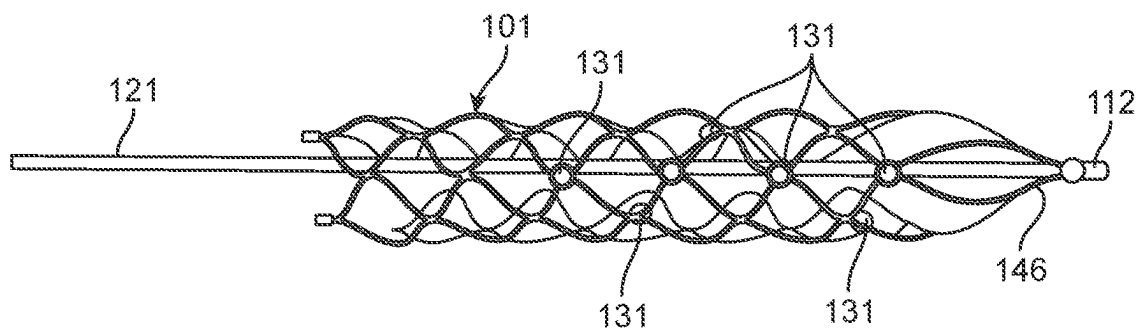


FIG. 38B

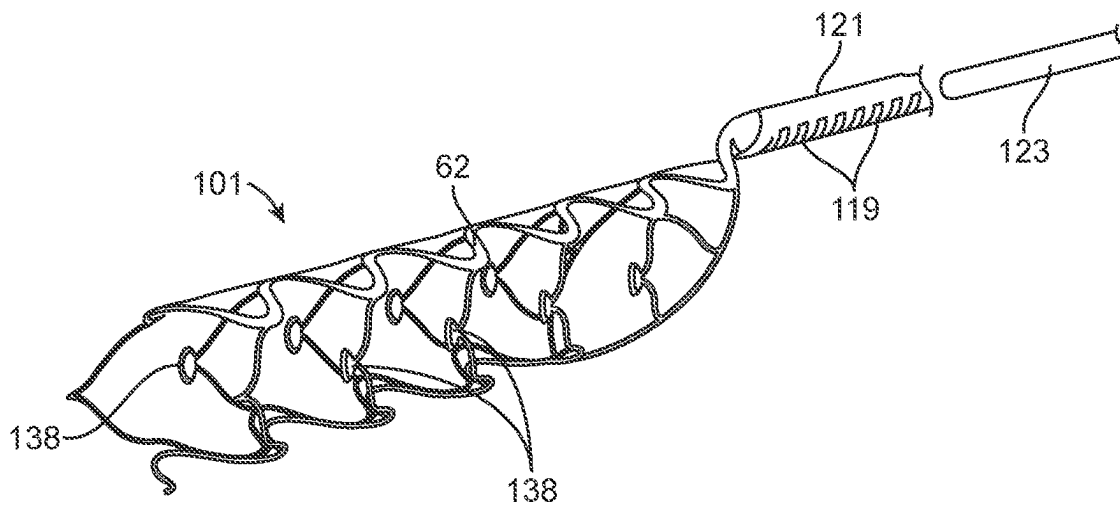


FIG. 38C

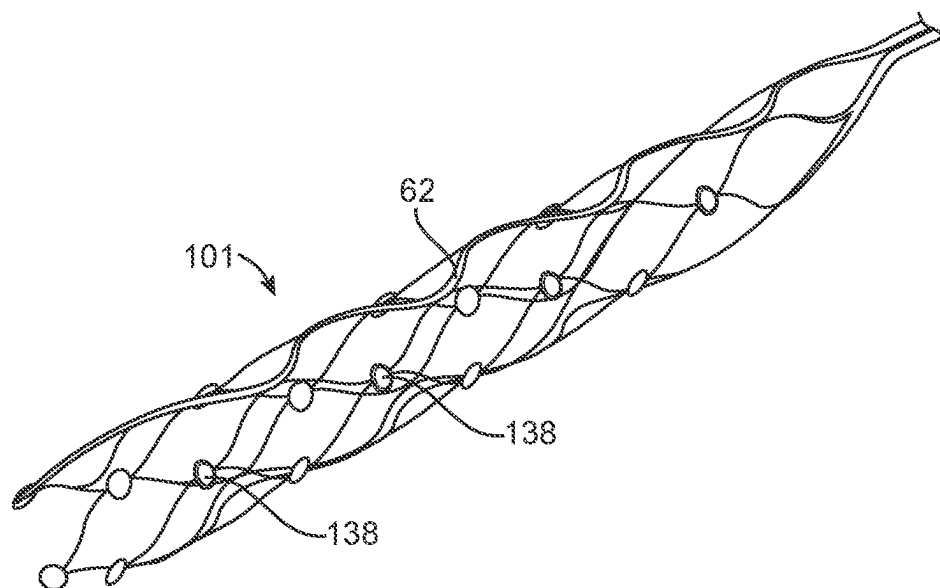


FIG. 38D

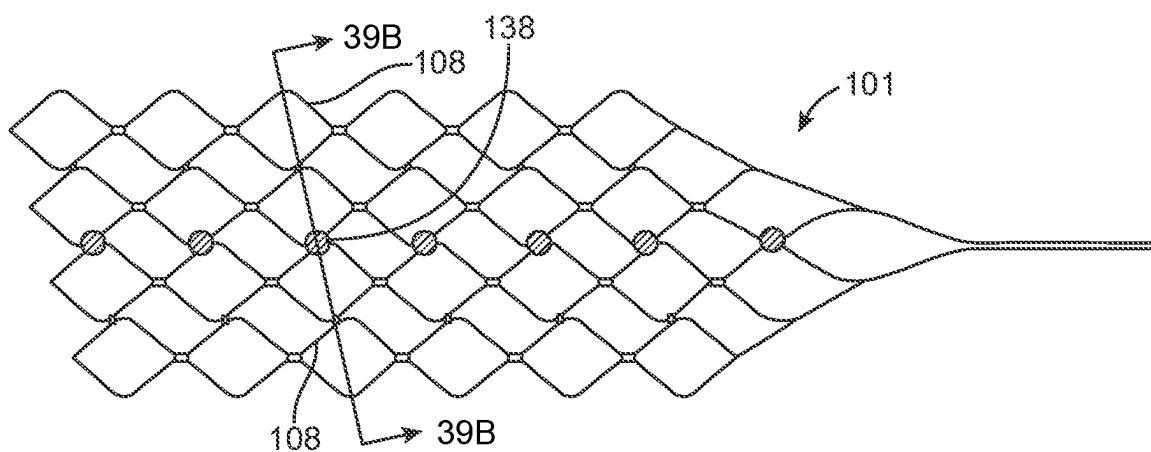


FIG. 39A

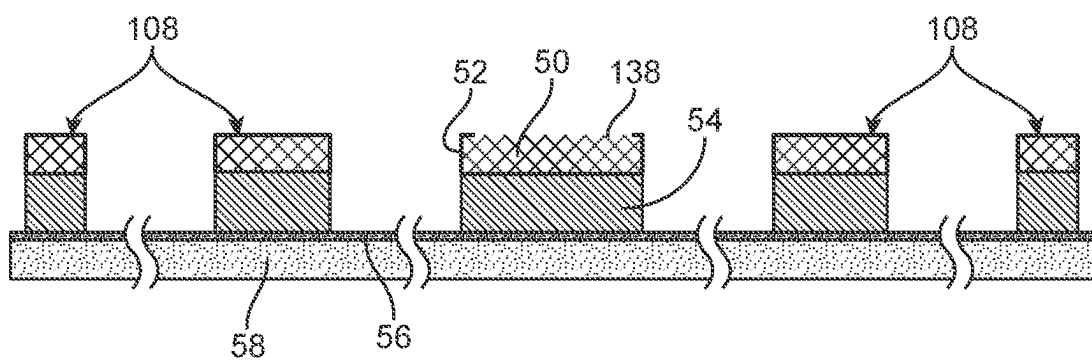


FIG. 39B

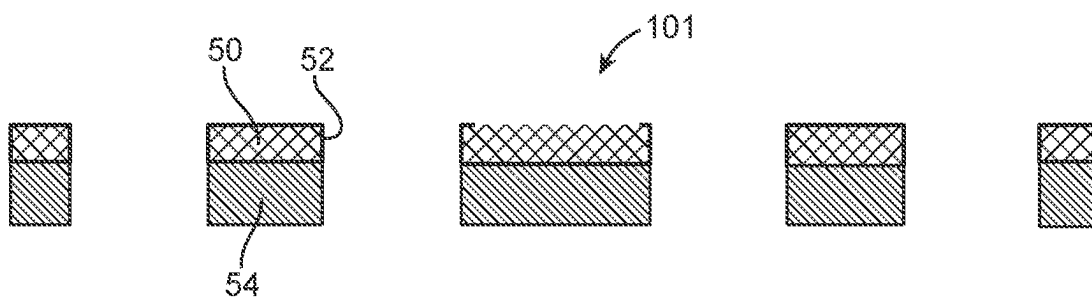


FIG. 39C

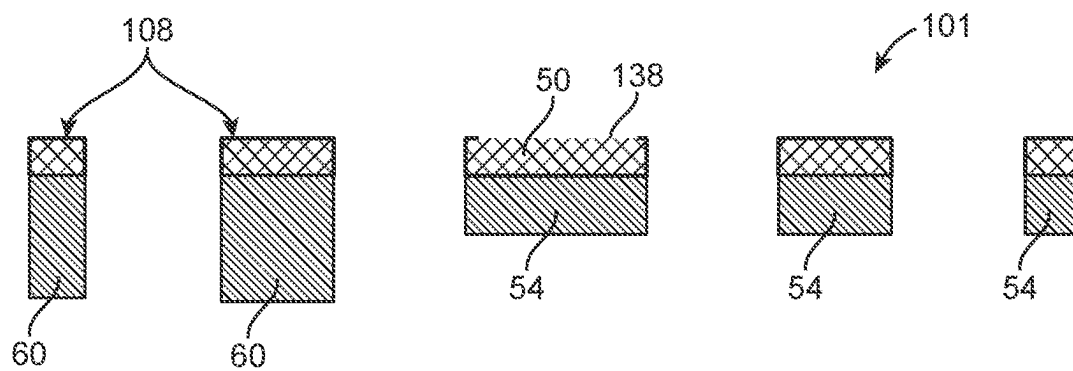


FIG. 40A

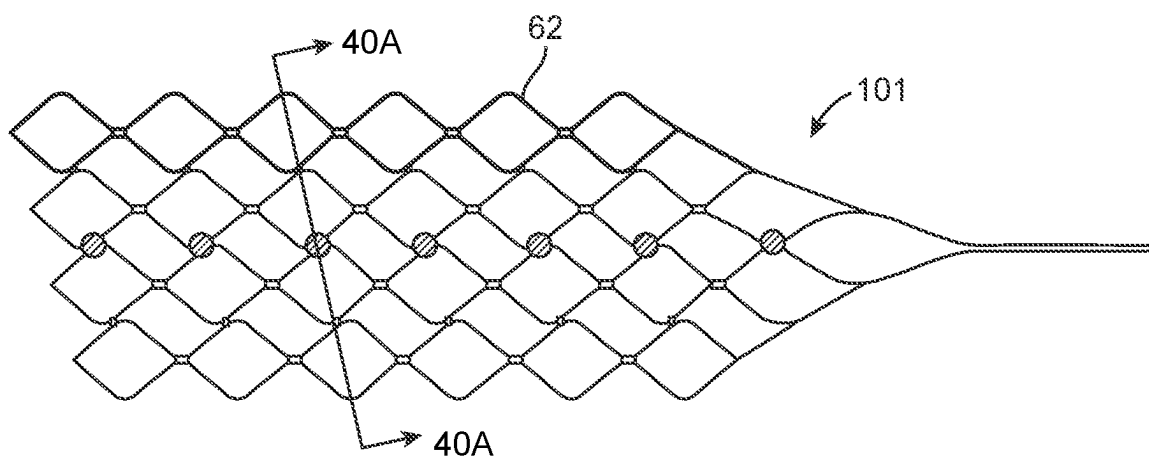


FIG. 40B

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MEDICAL DEVICE FOR SENSING AND OR STIMULATING TISSUE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 16/539,357, filed Aug. 13, 2019, which is a continuation of U.S. patent application Ser. No. 15/957,574, filed Apr. 19, 2018, now U.S. Pat. No. 10,485,968, which is a continuation of International Application No. PCT/US2016/057768, filed Oct. 19, 2016, now WO 2017/070252, which is a non-provisional application of Australian Provisional Application No. 2015904302 filed Oct. 20, 2015; Australian Provisional Application No. 2015905045 filed Dec. 4, 2015 and U.S. Provisional Application No. 62/379,625 filed Aug. 25, 2016, the entirety of each of which is incorporated by reference.

TECHNICAL FIELD OF THE INVENTION

The present invention relates to a medical device for implantation into a blood vessel of an animal.

BACKGROUND OF THE INVENTION

Any discussion of document, devices, acts or knowledge in this specification is included to explain the context of the invention. It should not be taken as an admission that any of the material forms a part of the prior art base or the common general knowledge in the relevant art in Australia or elsewhere on or before the priority date of the disclosure and broad consistency statements herein.

In the United States alone, nearly two million people suffer from various neuromuscular disorders where control of limbs is severely impaired. In many of these patients, however, the portion of the brain responsible for movement remains intact, and it is disease and trauma to the spinal cord, nerves and muscles that limit mobility, function and independence. For these people, the ability to restore lost control at even a rudimentary level could lead to a greatly improved quality of life.

At present, there are two primary options for restoring function. One option is to increase the capabilities of the remaining pathways, substituting paralysed or amputated muscles with those under voluntary control. While this method has been highly successful for amputees by re-innervating forearm nerves into abdominal muscles which control a bionic arm, the restored function greatly depends on the site of damage or condition, with people paralysed by brainstem or high cervical injuries only able to achieve minor functional improvement. A second option is to provide the brain with a new communication and control channel to convey messages to the external world. Currently, these brain controlled interfaces (BCIs) measure electroencephalographic or other electrophysiological activity via surgically implanted epidural, subdural, and intracortical electrodes. While cortical measurements performed with electrodes placed on the scalp enable non-invasive neuronal measurements, they require daily application and are prone to noise and movement related artefacts. Penetrating and non-penetrating intracranial electrodes, implanted after a craniotomy directly onto the surface of a cortical area, have much better signal to noise ratios (relative to scalp electrodes) and have been shown to enable rudimentary prosthetic hand operation. These methods, however, require invasive surgery and carry a relatively high risk of compli-

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cation, which can involve infections and bleeding. Furthermore, craniotomies are limited in access to the central nervous system, with many motor and sensory cortex areas hidden and inaccessible within cortical folds. These approaches are restricted in position and cannot be relocated once implanted and are subject to signal deterioration due to glial scar formation surrounding penetrating electrodes.

Thus, there remains a need to record and stimulate from cortical tissue in a method which is minimally invasive whilst also ensuring longevity and efficacy of recorded and induced signals.

By using blood vessels as a conduit to the brain, the risks associated with craniotomies, and the invasive creation of a burr hole in the skull of the patient is removed whilst also removing current noise and movement related artefacts observed with non-invasive scalp electrodes. Despite the minimally invasive benefits provided by these types of procedures, it is preferable that thrombus formation caused by the blockage of blood flow through a vessel is prevented. It is also preferable that the electrical energy delivered to the electrodes be as efficient as possible, which will reduce the burden placed on the electrical circuitry. Optimisation of wireless telemetry aimed to send power and data directly through the body to the implanted device, will enhance device functionality and negate the risk of infection caused through lead wires creating a direct passage between the vessel and the external environment. The ability to implant coils inside blood vessels will similarly reduce surgical risks associated with perforated vasculature.

Thus, there remains a need to provide improved intravascular electrodes, telemetry circuitry and implantation positions that are capable of more efficiently transmitting and receiving electrical energy between vessels and external circuitry, while minimizing the occlusion of blood flow.

It is generally desirable to overcome or ameliorate one or more of the above mentioned difficulties, or at least provide a useful alternative.

SUMMARY OF THE INVENTION

According to the present invention, there is provided a medical device for implantation into a blood vessel of an animal, including: (a) a stent movable between a collapsed condition of use for insertion into said vessel and an expanded condition of use for resiliently bearing against a wall of said vessel; (b) one or more electrodes coupled to the stent for stimulating and/or sensing activity of media proximal to the device, wherein the media includes tissue and/or fluid. The term stent is meant to include any support structure that maintains, carries, supports or incorporates the one or more electrodes within the tissue and/or fluid. The term stent can include conventionally designed medical stents, alternatively, the term stent can include any mechanical framework or scaffolding that positions electrode elements within a body lumen, such as a vessel, and facilitates electrical coupling of the electrode element(s) to a lead or other conductive structure. In certain variations, portions of the support structure itself can function as electrodes.

According to the present invention, there is also provided a method of recording of neural information or stimulation of neurons from the superior sagittal sinus or branching cortical veins of a patient using the above described device, including the steps of: (a) implanting the device in either the superior sagittal sinus or branching cortical veins; (b) receiving activity; and (c) generating data representing said activity; and (d) transmitting said data to a control unit.

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According to the present invention, there is also provided a method of for stimulation and recording neural information or stimulation of neurons from the visual cortex of a patient using the above-described device, including the steps of: (a) implanting the device in a vessel in the visual cortex of the patient; and (b) recording neural information associated with the vessel or stimulating neurons in accordance with received stimulation data.

According to the present invention, there is also provided a system for controlling use of apparatus coupled to an animal or human, including: (a) the above-described device, said device being adapted for placement within a vessel of an animal or human to stimulate and/or sense the activity of media proximal to the device; (b) a control unit adapted for communication with the device; (c) apparatus coupleable to the animal or human, said apparatus adapted for in communication with the control unit, wherein the control unit is adapted to perform the steps of: (i) receiving data from the device representing activity of media proximal to the device; (ii) generating control signals for the apparatus; and (iii) sending said control signals to said apparatus.

According to the present invention, there is also provided a control unit for controlling operation of apparatus coupled to an animal or a human, said control unit being adapted to perform the steps of: (a) receiving data from the above-described device, said data representing activity of media proximal to a vessel within which the device is placed; (b) generating control signals for controlling operation of the apparatus; and (c) sending said control signals to the apparatus.

The present disclosure further includes a medical device for use within a tubular body having a lumen, the medical device comprising: a frame structure forming a plurality of struts, where the frame structure is moveable between a reduce profile and an expanded profile in which a diameter of the frame structure increases; where at least one of the plurality of struts forming the frame structure comprises an electrically conductive material on a support material, the electrically conductive material extending along at least a portion of the strut and being covered with a non-conductive material; at least one electrode formed by an opening in the non-conductive material on the portion of the strut; and a lead located at an end of the frame structure and configured to be in electrical communication with the electrically conductive portion, the lead extending from the frame structure.

The medical device can further include a connector block configured to electrically couple the medical device to an external device, where the lead extends from the frame structure to the connector block.

In another variation, the present disclosure includes a method of recording of neural information or stimulation of neurons a patient the method comprising: receiving a signal representative of neural activity from a device positioned in a vessel of the patient; generating data representing said activity using the signal; and transmitting said data to a control unit; generating a control signal from the control unit; and transmitting the control signal to an apparatus coupled to the patient.

The present disclosure also includes a system for controlling an apparatus coupled to an animal or human. In one example, the system comprises a device adapted for placement within a vessel of the animal or human to stimulate and/or sense the activity of media proximal to the device; a control unit adapted for communication with the device, wherein the control unit is adapted to: (i) receive data from the device representing activity of media proximal to the

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device; (ii) generate a control signal; and (iii) transmit the control signal to said apparatus.

The system can include an apparatus selected from or more of the following: an exoskeleton; a prosthetic limb; a wheelchair; a computer; and/or an electrical or electro-mechanical device.

BRIEF DESCRIPTION OF THE DRAWINGS

Preferred embodiments of the present invention are hereafter described, by way of non-limiting example only, with reference to the accompanying drawing in which:

FIG. 1 is a diagrammatic illustration of a system for controlling use of apparatus coupled to an animal or a human;

FIG. 2A is a diagrammatic illustration showing parts of the system shown in FIG. 1;

FIG. 2B is a diagrammatic illustration showing of an additional variation of the system comprising two or more stents;

FIG. 3 is a diagrammatic illustration showing parts of the system shown in FIG. 1;

FIG. 4 is a diagrammatic illustration of a control unit of the system shown in FIG. 1;

FIG. 5A is a diagrammatic illustration of a medical device of the system shown in FIG. 1;

FIG. 5B is a cross-section view through the line A-A of the device shown in FIG. 5A;

FIG. 5C is a schematic diagram of a wireless chip;

FIG. 5D is a diagrammatic illustration of a medical device of the system shown in FIG. 1;

FIG. 6 is a diagrammatic illustration of a medical device located in a vessel;

FIGS. 7A to 7E are diagrammatic illustrations of medical device of the system shown in FIG. 1;

FIG. 8A is a diagrammatic illustration showing electrode mounting platforms

of a medical device of the system shown in FIG. 1;

FIG. 8B is a diagrammatic illustration showing placements of a medical device of the system shown in FIG. 1;

FIG. 9 shows diagrammatic illustrations of different electrode configurations;

FIG. 10 shows diagrammatic illustrations of different electrode configurations;

FIG. 11 is a diagrammatic illustration of a medical device of the system shown in FIG. 1;

FIG. 12 shows diagrammatic illustrations of different electrode configurations;

FIG. 13a is a diagrammatic illustration showing wire attachments to an electrode;

FIG. 13b is a diagrammatic illustration showing electrode lead wires wrapped around a shaft and covered in insulation forming a wire bundle or cable;

FIGS. 14 to 20 are diagrammatic illustrations showing different embodiments of the stent;

FIGS. 21a to 21c are diagrammatic illustrations showing deployment of different embodiments of the device;

FIGS. 21d and 21e show additional information regarding a helical lead 114;

FIGS. 22 to 24 are diagrammatic illustrations of a control unit of the system shown in FIG. 1;

FIGS. 25 and 26 are diagrammatic illustrations showing different stages of deployment of the device;

FIGS. 27 and 28 are diagrammatic illustrations of control units having ground electrodes attached thereto;

FIG. 29 is a diagrammatic illustration showing multiple vessels with multiple devices;

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FIG. 30 is a diagrammatic illustration showing a single vessel with multiple devices;

FIG. 31 is a diagrammatic illustration of a wireless electrode system;

FIG. 32 is a diagrammatic illustration of the system being used to record neural information or stimulation of neurons from the superior sagittal sinus (SSS) or branching cortical veins of a patient using the device;

FIG. 33 shows an image reconstruction of a human brain (eyes facing left) demonstrating superior sagittal sinus and branching cortical veins near the motor cortex (red) and sensory cortex (yellow);

FIG. 34 is a diagrammatic illustration showing a method for stimulation and recording neural information or stimulation of neurons from the visual cortex of a patient using the device;

FIG. 35 is a diagrammatic illustration showing vessels and muscles in a human arm;

FIG. 36 is an illustration of a human hand showing possible implant location to enable neural stimulation or measurement; and

FIG. 37 is a photo of a C-shaped ground electrode.

FIGS. 38A-38D illustrate examples of stents or scaffoldings having a plurality of electrodes disposed about the stent body.

FIGS. 39A-39C illustrate an example of integrated or embedded electrodes.

FIGS. 40A-40B show an example of a stent structure fabricated with dimensional variation to impart specific characteristics to the stent.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION

The system 10 shown in FIGS. 1 to 4 includes: 1) a medical device 100 designed for placement within a vessel 103 of an animal or human 110 to stimulate and/or sense the activity of media (tissue and fluids) proximal (adjacent or touching) to the device 100, whether this be located inside or outside the vessel 103; 2) a control unit 12 (also referred to as a connector block and telemetry system) adapted for communication with the device; 3) a communication conduit 14 for facilitating communications between the device 100 and the control unit 12; and 4) apparatus 16 coupleable to the animal or human 110, the apparatus 16 adapted for communication with the control unit.

The control unit 12 can be adapted to perform the steps of: (a) receiving data from the device 100 representing activity of media proximal to the device 100; (b) generating control signals for the apparatus 16; and (c) sending the control signals to the apparatus 16. In some variations, the system includes connector block (illustrated by element 12) that functions as connector and acts as an extension of the communication conduit. In variations of the system, the control unit/connector block: is hermetically sealed and insulates the leads from the device to the control unit; can be inserted using zero-contact force attachments or attachments that do not require excessive force to insert (i.e., using balseal spring contacts); has a portion of the lead that is made from a stiffer silicone or similar material for handling and insertion into the connector. Variations of the device can include markers to identify portions of the leads that are stiffer (and can be handled) to distinguish from leads that cannot be handled. Such markers can include line-style markers, different colours or other indicators to clearly identify the regions. Variations of the connector block can have a fitting (e.g., clasp) such that multiple connectors can

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be inserted (i.e., two contact connectors (with 8 contacts each) for a 16 electrode Stentrode lead). The fitting can ensure securing of the contacts, alignment and prevention of water ingress

When the medical device 100 is inserted adjacent to the motor cortex in the manner shown in FIGS. 2A, 2B, and 3, the system 10 can be used, for example, to control operation of an exoskeleton, and/or an artificial limb in the manner shown in FIG. 1.

This device 100 is implanted into blood vessels 103, from which, it will utilise electrodes mounted on a self-expanding member 101 to record or stimulate neighbouring tissue. Information is to be passed from or to the electrodes through the communication conduit 14, inside of the blood vessel 103, to a telemetry system 12 that, in turn, passes information (using wires or wirelessly) to or from an external apparatus 16, which includes (but is not limited to) one or more of the following:

(a) an exoskeleton; (b) wheelchair; (c) computer; and/or (d) other electrical or electro-mechanical device.

As such, in one specific application, the implanted medical device 100 has the capability to enable a paralysed patient 110 to use their thoughts directly to command and control a gait aid such as an exoskeleton or robotic legs 16.

Other applications for the implantable medical device 100 include (but are not limited to): (a) detection and prevention of seizures; (b) detection and prevention of involuntary muscular or neural control (for example to alleviate symptoms associated with: (i) multiple sclerosis; (ii) muscular dystrophy; (iii) cerebral palsy; (iv) paralysis and (v) Parkinson's); (c) detection and therapeutic alleviation of neurological conditions, such as: (i) post-traumatic stress disorder; (ii) obsessive compulsive disorder; (iii) depression; and (iv) obesity; (d) direct brain control of computers and equipment, such as: (i) vehicles; (ii) wheelchairs; (iii) gait aids; robotic limbs; (e) direct input for sensory stimulation for: (i) blindness (connection to a camera); (ii) deafness (connection to microphone); (iii) proprioception (connection to touch-sensitive robotic and computer systems); (f) internal assessment of personal health and wellbeing: (i) heart rate; (ii) respiration rate; (iii) temperature; (iv) environmental conditions; (v) blood sugar levels; and (vi) other biochemical and neurological markers; (g) internal communication (telepathy) between implanted groups of people utilising the device for information transmission, auditory, visual and proprioceptive feedback (for example, real time communication of what the implantee sees or hears); and (h) augmentation and optimisation of musculoskeletal control and dexterity (for performance enhancement or rehabilitation).

FIG. 2B illustrates a two-stent 101 system. For purposes of illustration, the stents are positioned in a single vessel. However, the stents can be configured such that they can be positioned in separate vessels. The stents 101 can be joined by non-conductive material to form a power receiver and transmitting antenna. Alternatively, the stents can be coupled by one or more wires or conductive elements. Moreover, the system can include active electronics between the stents 101.

The devices described herein can be positioned in any number of areas of brain structures depending upon the desired outcome. For example, as discussed in Teplitzky, Benjamin A, et al. "Computational modeling of an endovascular approach to deep brain stimulation." *Journal of Neural Engineering* 11.2 (2014): 026011. stents can be positioned as follows: Internal capsule for depression and obsessive compulsive disorder (OCD); thalamus for epilepsy (E), Parkinson's Disease, essential tremor, Tourette syndrome,

consciousness disorder, chronic pain, obsessive compulsive behavior; fornix for Alzheimer's disease; globus pallidus internus for dystonia, depression, Tourette syndrome; hippocampus for epilepsy; hypothalamus for obesity, anorexia mentosa; inferior thalamic p'duncle for depression and obsessive compulsive disorder; lateral habenula for depression, obesity, anorexia mentosa; nucleus accumbens for depression, obsessive compulsive disorder, addiction, obesity, anorexia mentosa; periaqueductal/periventricular for chronic pain; subgenal cingulate white matter for depression; subthalamic nucleus for Parkinson's Disease, dystonia, depression, obsessive compulsive disorder, epilepsy; and ventral capsule for obsessive compulsive disorder.

1. Medical Device

As shown in FIGS. 5a, 5b, 5d and 6, the medical device 100 generally includes: a. a collapsible and expandable stent 101; b. a plurality of electrodes 131 coupled to the stent 101; c. electrode lead wires 141 electrically coupled to electrodes 131; d. an olive 112 coupled to the stent 101 by an olive wire 114 for preventing perforation of vessels during implantation; e. implanted chips; f. contacts 151 couple to the lead wires 141 to enable communication between the device 100 to the control unit 12; and g. a stent shaft 121 is used to deploy the device 100.

Electrode lead wires 141 can be electrically connected to at least one electrode and will be wound around the stent strut lattice 108 such that mechanical compression and extension is not interfered with. Electrode wires 141 may be wound around the stent shaft 121, thread through a stylet shaft or may form part of the stent shaft directly. Lead wires 141 will form connections with electrode contacts 151 on the opposite end of the stent shaft to the stent, whereby electrical contact a connector block mechanism 12 enables the connection path with external equipment 16, which included but is not limited to computers, wheelchairs, exoskeletons, robotic prosthesis, cameras, vehicles and other electrical stimulation, diagnostic and measurement hardware and software.

The term electrode 131 is used in this specification to refer to any electrical conductor used to make contact with media in and/or around a blood vessel 103.

A detailed description of the operation of each of these components is set out below.

The Stent

The stent 101 includes a plurality of struts 108 coupled together with strut cross links 109.

In the arrangement shown in FIG. 7a, the device 100 includes nine electrodes coupled to the stent 101 in a linear pattern. As shown, the stent 101 appears flat. The top of the stent 101 may be directly joined to the bottom of the stent 101 or will curve around to meet (without permanent attachment) the bottom of the stent 101.

Alternatively, the device 100 includes a stent with any suitable number of electrodes 131 arranged in any suitable configuration. For example, the electrodes can be configured as follows: the sinusoidal arrangement of electrodes 131 shown in FIG. 7b; the spiral arrangement of electrodes 131 shown in FIG. 7c to enable 360 degree contact of an electrode to the vessel wall once deployed; the reduced amplitude sinusoidal arrangement of electrodes 131 shown in FIG. 7d for increased coverage whilst still ensuring only one stent is at each vertical segment; and the dense arrangement of electrodes shown in FIG. 7e for increased coverage. The stent 101 is laser cut or woven in a manner such that there is additional material or markers where the electrodes 131 are to be placed to assist with attachment of electrodes and uniformity of electrode locations. For example, if a stent

101 was fabricated by laser cutting material away from a cylindrical tube (original form of stent), and, for example, electrodes are to be located at 5 mm intervals on the one axis, then electrode mounting platforms 107, 108 can be created by not cutting these areas from the tube. Similarly, if the stent is made by wire wrapping, then additional material 107, 108 can be welded or attached to the stent wires providing a platform on which to attach the electrodes. Alternatively, stents can be manufactured using thin-film technology, whereby material (Nitinol and or platinum and or other materials or combinations of) is deposited in specific locations to grow or build a stent structure and/or electrode array.

Electrodes As particularly shown in FIGS. 8a, the device 100 includes electrode placements 107 coupled to strut cross links 109. The placements 107 are used to coupled the electrodes 131 to the stent. An alternative embodiment of the placements 106 is shown in FIG. 8b. In this embodiment, the placements are circular.

As shown, the electrodes 131 are located on or at the stent cross links 109. Locating the electrodes in these positions allows for changes in shape of the stent 101 (i.e expanding and collapsing) without significantly affecting the integrity of the electrodes. Alternatively, may also be located in between the stent strut crosslinks (not depicted).

FIG. 9 depicts different electrode geometries which include but are not limited to: flat discs 161; cylinders or rings 162; half-cylinders or rings 163; spheres, domes or hemispheres 164; hyperbolic paraboloids 165; and double electrodes or electrodes whereby they are preferentially longer along one axis 166.

As shown in FIG. 10, the electrodes 131 preferably include shape memory material and hence the electrodes 131 may be uninsulated sections of the device 100. As shown, the electrode 131 inside a patient and the vessel 104 is unobstructed. After activation of shape memory, the electrode 131 conforms to better fit the vessel wall 103.

To enhance contact and functionality of the device 100, electrodes 131 include the attachment of additional material (shape memory alloy or other conducting material) through soldering, welding, chemical deposition and other attachment methods to the stent 101 including but not limited to: directly on or between the stent struts 108; to lead wires 14 passing from the electrodes 131 to wireless telemetry links or circuitry; and directly to an olive 112 placed on the distal aspect of the device 100 to or stent shafts.

To further enhance the device 100 performance, there may be one or more electrodes 131 per wire strand 141 and there may be one or more strands 141 utilised per device 100. These strands 141 may be grouped to form a bundle 144, which may be woven in alternate sinusoidal paths around the stent struts 108 in the manner shown in FIG. 11. Similarly, there may be one or more wires 141 designated to each electrode 131 and hence there may be one or more electrodes 131 per device 100. Thus, multiple electrodes 131 may be used simultaneously.

To optimise the ability of the electrodes 131 to stimulate or record from medium (including but not limited to neural tissue, vascular tissue, blood, bone, muscle, cerebrospinal fluid), the electrodes 131 may be positioned at pre-determined intervals based on the diameter of the target vessel 103 to allow each of the electrodes 131 to be in contact with the vessel 103 in the same orientation (ie, all electrodes facing to and in contact with the left vessel wall upon deposition). Electrodes 131 may be mounted such that recordings or stimulation can be directed to all 360 degrees

of the vessel simultaneously. Similarly, to enhance the recording and stimulation parameters of the electrodes **131**, the electrode sizes may be varied, with larger electrodes **131** used to assess greater areas of neighbouring medium with smaller electrodes **131** utilised for localisation specificity.

Alternatively, the electrodes **131** are made from electrically conductive material and attached to one or more stents, which form the device **100** and allow for multiple positions. In this embodiment, the electrodes **131** are made from common electrically active materials such as platinum, platinum-iridium, nickel-cobalt alloys, or gold, and may be attached by soldering, welding, chemical deposition and other attachment methods to one or more lead wires **141**, which may be directly attached to the shape memory shaft(s). The electrodes **131** are preferably one or more exposed sections on the insulated lead wire **141** and the electrode lead wires may be wrapped around one or more shape memory backbones. There may be one or more electrodes and lead wires wrapped around a single shape memory backbone, and, where multiple shape memory backbones are used in the one device, the backbones may have different initial insertion and secondary deposition positions. Thus, they may be used for targeting multiple vessels simultaneously.

As shown in FIG. **12**, the electrodes **131** can be designed such that they are carriers of substances **134** and solutions such as therapeutic drugs, including but not limited to anti-thrombogenic, and materials. In this embodiment, the electrodes **131** are designed to release the drugs, either passively through diffusion or through control by an implanted electrical clock or manually through electrical stimulation of the electrodes **131**. In this embodiment, the electrodes **131** are made from materials that have portions of the electrodes **131** that are not electrically conductive.

The drug **134** is preferably released into the vessel **104** upon timed, natural, electrical or otherwise activation, or into the vessel wall **103**.

Electrode Wires

The electrode wires **141** are electrically coupled to respective electrodes in the manner shown in FIG. **13a**. As shown, the electrical attachment **135** and the back face of the electrode is covered in a non-conductive substance **136**.

The lead wires **141** are preferably wrapped around the stent **101** and along a shaft **121**.

As shown in FIGS. **5a**, **5b** and **13b**, the electrode lead wires **141** are wrapped around the shaft **121** and covered in insulation **122** forming a wire bundle or cable. A sleeve **153** wraps around the wire bundle at the location of the contact **151**, whereby at least one wire **141** is wrapped around the sleeve **153** and connected to the contact **151** at a connection weld point **152**. The over-molding **154** ensures a uniform diameter is present between contacts.

The sleeve **153** covers the wire bundle **142** with an exposed section of wire **141** attached **152** to a contact **151**.

Distal electrodes and/or markers and/or buffers are also depicted **112** attached via a wire **114** to the stent **101**. The shaft **121** is attached at the end of the stent at the attachment/detachment zone **115** and is shown passing through the sleeve **142** and electrode contacts **151** to exit behind past the connector securement point **155**.

The lead wires **141** shown to be inside the sleeve **142** where they are wrapped around the shaft **121** where they make electrical contact at a contact weld **152** to the electrode contacts **151**. An overcoat **154** is shown to ensure uniform diameter of the device between the contacts. The shaft **121** may be detached at the detachment zone **115** and removed following deployment in a vessel.

As shown in FIG. **13b**, lead wires **141** are connected to electrode contacts **151**. Electrode lead wires **141** are initially wrapped around a shaft **121** covered in insulation **122** forming a wire bundle or cable. A sleeve **153** is placed around the wire bundle at the location of the contact, whereby at least one wire **141** is wrapped around the sleeve and connected to the contact **151** at a connection weld point **152**. Over-molding **154** may be used to ensure a uniform diameter is present between contacts.

As particularly shown in FIG. **5b**, the stent shaft **121** is coated in an insulative layer **122**, has a plurality of wires **141** that are insulated **143** and grouped in an insulated bundle **142** wrapped around it. A sleeve **153** covers the wire bundle **142** with an exposed section of wire **141** attached **152** to a contact **141**.

The wires **141** are made from electrically conductive materials including but not limited to Platinum, Platinum/Tungsten, Stainless Steel, Nitinol, Platinum/Iridium, Nickel-Cobalt Alloys, or other conductive and biocompatible materials.

The wires **141** are between 10 um and 100 um thick (diameter), stranded cable or monofilament, and connect the electrodes **131** to the contacts **151**. Alternatively, the wires **141** connect the electrode **131** to wireless circuitry retained on the stent or shaft.

The wires **141** are insulated with non-conductive material (ie, Teflon or polyimide). The wires **141** are wrapped around the stent struts in a sinusoidal pattern as shown in FIG. **11**. Alternatively, the wires **141** are wrapped in a helical tube or wire bundle or cable, with the wire or bundle between 300 um and 2 mm in diameter (thickness).

The wires **141** are connected to contacts **151** using wire wrapping, conductive epoxy, welding, or other electrically conductive adhesion or connection means.

Olive

In the embodiment shown in FIG. **5a**, the device **100** includes an olive **112** mounted at the distal tip to reduce risk of perforation and to improve device **100** safety during the implantation and deposition phase. In this arrangement, the olive **112** is directly connected to the front of the device **100** and act as a buffer, which is the first aspect of the device that comes in contact with the deployment catheter or the vessel during deployment. The olive **112** can additionally be used as a radiopaque distal marker. The olive **112** can be configured and attached to the stent **101** in many different forms including, but not limited to, the following:

i. Flexible Cord

As shown in FIG. **5a**, the olive **112** is placed at a distance from the front of the stent **101**, connecting with the stent **101** via a flexible cord **114**.

ii. Spring Olive

FIG. **14** depicts an olive placed on the distal end of a stent **101** whereby the olive is comprised of a buffer which may or may not be electrically active and function as an electrode **112** connected to the stent **101** by a flexible spring or helically wound wire **111**.

iii. Multiple Olives

FIG. **15** depicts a plurality of olives placed on the distal end of a stent **101** whereby the olive is comprised of a plurality of buffers which may or may not be electrically active and function as an electrode **113**.

iv. Short Olive

FIG. **16** depicts an olive placed on the distal end of a stent **101** whereby the olive is connected directly to the end of the stent which may or may not be electrically active and function as an electrode **112**.

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v. Shaped Wire Olive

FIG. 17 depicts an olive placed on the distal end of a stent **101** whereby the olive is a flexible wire which may or may not be electrically active and function as an electrode and may or may not be shaped as a shepherd's crook **114**.

vi. Wire Olive

FIG. 18 depicts an olive placed on the distal end of a stent **101** whereby the olive is comprised of a buffer which may or may not be electrically active and function as an electrode **112** connected to the stent **101** by a flexible wire **114**.

vii. Olive with Detachment Zone

FIG. 19 depicts an olive placed on the distal end of a stent **101** whereby the olive is comprised of a buffer which may or may not be electrically active and function as an electrode **112** connected to the stent **101** by a flexible wire **114**. This figure further depicts a shaft **121** that is connected to the stent **101** via an attachment and/or detachment zone **115**.

FIG. 20 further depicts the shaft **121** that is detached from the stent **101** via the attachment and/or detachment zone **115**.

The flexible wire **114** includes but is not limited to electrically conductive and electrically insulating wires, springs, helical leads and tubes which may have a buffer at the front. Alternatively, the buffer is electrically conductive and acts as an electrode, comprising all the features of stent-mounted electrodes.

Implanted Chips

Implanted electrical circuitry (chips) are preferably used to control the stimulation and measurement of the electrodes **131**. The chip can be implanted in place of an electrode (or elsewhere mounted on the stent), where the chip has the capacity to transmit the signals. The chip includes circuitry for: (a) signal amplification; (b) signal multiplexing; and (c) transmission of power and data.

The electrodes **131** are attached to one or more electrical chips (whereby the chip is defined as the electrical circuitry as well as the substrate which the chip is built on). Miniaturised chips are mounted on the stent **101** in a similar manner and position to the electrodes **131**.

Alternatively, these chips may be attached at a distance from the neural recording or stimulation site such as the neck or pectoral region, or the chip may connect directly to external hardware, such as current sources, recording equipment or prostheses.

The chips preferably include circuitry for stimulation of neural tissue (current and/or voltage sources, batteries and/or capacitors or charge/energy storing components and switch matrices, etc) and circuitry for the recording of neural activity (amplifiers, power sources, switch matrices, etc) and blood composition (such as pH meters, salts and saline composition, glucose etc).

Further, chips may have circuitry required for the transmission of power and data through telemetry coils and self-monitoring hardware such as thermal sensors.

The depiction of the wireless chip **195** shown in FIG. 5c, whereby the microprocessor **191** is shown as well as other components **193** (eg, capacitors, multiplexors, clocks, wireless transmitters, receivers etc). This depiction has two coils that can be used for transmission and receiving of both power and data, shown as a large coil **192** and a small coil **194**.

The chip itself may contain a telemetry coil for the transmitting and receiving power and data and may contain a magnet to enable alignment with adjacent chips and telemetry coils or may be attached to shape memory alloys or other materials in which the telemetry coils are comprised.

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The chip is preferably flexible, and may be pre-curved to the diameter of the vessel to allow for the deposition of the chip within a vessel. Thus, the chip may contain shape memory alloys or polymers to conform the chip to the curvature of the vessel during the deposition phase. The chip may also be mounted on a bioabsorbable or biodegradable substrate to allow for integration within a vessel. Multiple chips may be used simultaneously.

f. Contacts

As particularly shown in FIGS. 5a and 5b, electrode contacts **151** are required to enable connection of the device **100** to external equipment in the situation where wireless circuitry is not employed. The electrode contacts **151** are preferably made from materials similar to those used by the electrodes and will be of similar diameters. The contacts **151** are electrically insulated from each other and will be connected to the electrode lead wires **141** by (but not limited to) conductive epoxy, laser or resistance welding, soldering, crimping and/or wire wrapping.

The contacts **151** are platinum rings or rings of other conductive, biocompatible materials. The contacts can be made from or contain magnetic materials (ie, Neodinium).

The contacts **151** are preferably: (a) between 500 um and 2 mm in diameter; (b) between 500 um and 5 mm in length; and (c) between 10 um and 100 um in thickness.

The contacts **151** are shaped as discs, tubes, paraboloids or other shapes similar to those used for the electrodes **131**.

The contacts are placed over non-conducting sleeve (including but not limited to a silicone tube, heat shrink, polymer coating) to assist with electrical insulation of other lead wires and electrode and stent wire, and to assist in retaining shape tubular shape whilst allowing some flexibility.

Preferably, the contacts **151** have a contact to contact separation of between 100 um and 10 mm.

The contacts **151** are formed through wire wrapping of the wires **141**.

Preferably, at least one contact **151** is a dummy connector (including but not limited to a metal ring, magnetic ring, plastic tube). A dummy connector in this instance is a connector that is not in electrical contact with an electrode, instead, the purpose is to enable a connection or securing point (ie, through a screw terminal) to the device in a desired location and such that the contacts (connected to electrodes) are not damaged.

The contacts **151** are separated by a non-conductive sleeve (including but not limited to a silicone tube, heat shrink, polymer coating) to reduce electrical noise and prevent contact between superficial lead wires **141**.

g. Shaft

As shown in FIG. 21a, to enable deployment, a flexible shaft **121** is connected to the device **100**. In the example shown in FIG. 21a, the shaft **121** is connected at the distal end of the device **100** such that it acts to pull the device **100** from the front.

In the alternate embodiment shown in FIG. 21b, the shaft **121** is attached to the proximal end of the device **100** such that the shaft **121** pushes the device **100** from the back of the stent **101**. In this embodiment, medical device **100** includes a plurality of electrodes **131** mounted to a stent **101** with electrode lead wires **141** wrapped around the stent **101** and the shaft **121** and covered in a sleeve **142**. Distal electrodes and/or markers and/or buffers are also depicted **113** as is the stent detachment zone **105**.

The further embodiment shown in FIG. 21c includes a double tapered stent **101** with mounted electrodes **131** and a stent shaft **121** attached to the stent **101** at the stent attach-

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ment/detachment zone **105**. Another attachment/detachment zone **115** at the front of the stent **101** connects the stent **101** to the olive wire **114** and a stylet sleeve **124**, through which, a removable stylet **123** is placed. Electrode wires **141** are shown as wrapped around the outside of the stylet sleeve **123** or as being fed through the centre.

There may be a plurality of wires, with both pushing and pulling abilities. The stent shafts **121** may be implanted permanently or may be designed to be detached and removed. In this embodiment, the attachment/detachment zone will be located at the junction of the stent shaft **121** and the stent **101**. Detachment methods include, but are not limited to, electrochemical detachment, mechanical detachment and thermo-electrical detachment.

The stent shaft **121** can be used as a backbone for electrode lead wires **141**, assisting the stability of the electrode lead wires **141** as they traverse from the electrodes **131** to the electrode contacts. In this embodiment, the electrode wires **141** are in a polymer **142**, (including but not limited to shrink wrap, heat shrink, parylene, silicone, Teflon, etc) to provide additional mechanical support, assist in water retention and to enable coatings to be deposited onto the stent shaft where wires are present.

The stent shaft **121** may be a stylet that is removed following implantation and deposition of the device **100**. In this embodiment, the stent shaft **121** may be a cylindrical tube such that the stylet **123** can be fed through the centre of the tube **121**.

The wires **141** are preferably thread through the middle of a stylet sleeve.

Preferably, the wires **141** are wrapped around the stent shaft or stylet sleeve.

In a further embodiment, the electrode wires **141** that connect the electrodes **131** to the contacts **152** are wrapped in a wire bundle **144** and wrapped around an internal lumen tubing **145** in a helical form such that there is an internal lumen **147** whereby a removable stylet **148** can be thread during insertion and removed following deployment. This embodiment enabled removability of the stylet **148** and flexibility of the wire bundle **144** that is over coated in an external tubing **146**.

FIGS. **21d** and **21e** show additional information regarding a helical lead **114**. As depicted, the helical **114** lead includes wire bundle **144** wrapped around an internal lumen tube **145**. Through the internal lumen **147**, a removable stylet **148** can be thread during delivery and removed following device placement

Control Unit

The control unit **12** shown in FIG. **2** is a wireless controller, relaying information and power through the skin wirelessly.

The connector block **12** in FIGS. **22**, **23**, and **24** are passive devices (ie, no circuitry). Essentially, it functions as an intermediate connection between the device **100** and external equipment. The device **100** is inserted into the connector block **12** whereby the device **100** contacts make electrical contact with internal contacts contained within the connector block **12**. These internal contacts of the connector block **12** then form a thicker wire bundle which passes through the skin (the rest of the connector block is implanted) and can be connected to external equipment.

Essentially, as we are limited in space (the entire device must pass through a catheter as the catheter needs to be removed over the device after implantation) the connector block enables attachment of larger items to the thin device **100**.

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The embodiments shown in FIGS. **22**, **23** and **24** are the same, although only FIG. **24** shows the wire that goes through the skin.

The control unit **12** shown in FIG. **22** is shaped to receive and make electrical connection with the lead **14**. The control unit include contacts rings mounted on the inside. Here, the connector block **12** is secured and ensured water-tight through attachment of silicone and/or sutures at the grooved end.

The wireless system that is implanted on the stent directly is essentially the same (although a miniaturised version) of the wireless system **12** in FIG. **2**.

As shown in FIG. **23**, the electrode lead **14** is inserted and a silicone gasket is used to make a watertight seal following

FIG. **24** depicts a connector block whereby the electrode lead **14** is thread through the connection opening **172** whereby the contacts connect with the electrically conductive connectors **175** inside the connector block body **173**. Separation and electrical insulation and water-tightness is increased through silicone (or otherwise) separators **174**. Contacts **175** are welded (or otherwise) to connector block wires **179** that may form a silicone or otherwise **181** encased bundle **181** to terminate at a wireless or direct electrical connection port **183**.

Method of Using the System

The device **100** is movable between an insertion position shown in FIG. **25** and the deposition or scaffolding position shown in FIG. **26**.

In the insertion position, the device **100** is contracted and thus thin enough to be threaded through the vasculature pathway from within a catheter from an entry point (ie, the jugular vein) to a deposition point (eg, the motor cortex).

When arranged in the deposition or scaffolding position, the device **100** is in an expanded condition where scaffold electrodes mounted on the outside of the stent **101** as pressed against the vessel wall. This expanded position anchors the device **100** in its location within the vessel **103**. Further, this deposition position is designed such that it has a minimal effect on blood flow integrity through the vessel **103** in which the device **100** is deposited. The scaffolding position may be synonymous to a spring, coil or helical strand, whereby the device **100** is in contact with the vessel wall only, reducing the effect on blood flow. Electrodes **131** may also be mounted on the inside of the stent **101** such that information from fluid flowing through the expanded stent **101** can be measured. For a stent **101** to be removed or relocated, additional shafts (other than that used for initial deployment) are required. These are explained in the context of this invention, with both single tapered and double tapered designs used.

To enable the device **100** to be arranged in multiple positions, the material used is such that multiple states are possible. These materials include, but are not limited to, Nitinol and other shape memory alloys and polymers. Further, to enhance the long term biocompatibility of the device **100**, the polymers may be bioabsorbable or biodegradable, with a time of degradation similar to the time in which fibrosis occurs over the device **100**. Hence, the electrodes **131** (which preferably are not designed to degrade, and may be made from Nitinol, shape memory alloys, conductive polymers, other non-shape memory alloys and inert and biocompatible metals such as platinum, iridium, stainless steel and gold) will be all that remains of the initial device **100** and will become embed inside the blood vessel **103**, further enhancing the stability of the device **100** at the location of deposition

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Device in Blood Vessel (After Deployment)

FIG. 6 depicts a medical device 100 in the expanded or deposition or scaffolding position. The device 100 includes a stent 101, distal olives and/or proximity markers 112, a wire 114 attaching the stent 101 to the olive 112, a plurality of electrodes 131, and an attachment/detachment zone 115 whereby the shaft is connected to the stent 101 having been deployed in a blood vessel 104. Stent 101 mounted electrodes 131 are in direct apposition with the vessel wall 131 and are depicted as not interruptive of blood flow to any vessel (both the vessel the device is deployed in and other connected vessels). Here, the olive 112 can be used to direct the medical device into the desired vessel 104.

Device in Blood Vessel Pre-Deployment

FIG. 25 depicts a medical device 100 during implantation (surgical deployment phase) as it is being thread through vessels 104 inside a catheter 102. The stent 101, electrodes 131, stent detachment zone 105 and stent distal markers/electrodes/buffers 113 are shown, as are the vessel walls 103. Here, the catheter 102 is being used to select and direct the device into the desired vessel 104.

Device in Blood Vessel After Deployment

FIG. 26 depicts a medical device 100 in the expanded or deposition or scaffolding position comprising a stent 101, distal olives and/or proximity markers 113, a plurality of electrodes 131, lead wires 141 and a stent detachment zone 105 being deployed in a blood vessel 104 through a deposition catheter 102. Stent 101 mounted electrodes 131 are in direct apposition with the vessel wall 103 and are depicted as not interruptive of blood flow to any vessel (both the vessel the device is deployed in and other connected vessels).

Ground Electrode

The system preferably includes a ground electrode 167, configured in the manner shown in FIG. 27, which is used to assist and improve the quality of the recorded signals or to provide an electrical return path for stimulation applications. Here the ground electrode may be placed on the connector block provided it is implanted. Ground electrode 167 is preferably directly attached to the outside of the wireless controller 12.

An alternative embodiment of the ground electrode 167 is shown in FIG. 28. Ground electrode 167 on the outside of the controller 12.

The platinum C-shaped ground electrode 167 shown in FIG. 37 is embed in silicone 181 with a red helical lead wire 141 that is attached to a standard electrical terminal 169. Dacron mesh is used to assist secure the electrode and wire to tissue.

FIG. 29 shows a vessel with multiple devices 100 inserted in different vessels 104 to access different areas.

FIG. 30 shows a single vessel 104 with multiple devices 100 implanted to cover a larger area.

FIG. 31 a wireless electrode system 1000 showing electrodes mounted on a stent 101 within a blood vessel 104 overlying the motor cortex in a human that are picking up neural information and relaying this information to a wireless transmitter 1002 located on the stent 101. Note the stent 101 has been deployed and the stylet has been removed (ie, only the stent 101, electrodes, electrode wires and wireless system 1002 remains). The information is wirelessly transmitted through the skull to a wireless receiver 1004 placed on the head, which in turn, decodes and transmits the acquired neural information to a prosthetic limb 16.

As shown in FIG. 32, the device 100 can be used to record neural information or stimulation of neurons from the superior sagittal sinus (SSS) or branching cortical veins of a

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patient using the device 100, including the steps of: (a) implanting the device in either the superior sagittal sinus or branching cortical veins; (b) receiving activity; and (c) generating data representing said activity; and (d) transmitting said data to a control unit. Stent 101 implanted in SSS over motor cortex acquiring (i.e. receives) signals that are fed through the wire to external equipment 12.

FIG. 33 shows an image reconstruction of a human brain (eyes facing left) demonstrating superior sagittal sinus and branching cortical veins near the motor cortex (red) and sensory cortex (yellow)

FIG. 34 shows a method of for stimulation and recording neural information or stimulation of neurons from the visual cortex of a patient using the device 100, including the steps of: (a) implanting the device in a vessel in the visual cortex of the patient; and (b) recording neural information associated with the vessel or stimulating neurons in accordance with received stimulation data.

As particularly shown in FIG. 35, the device 100 is delivered through a vessel 104 deposited in a muscle for direct muscular stimulation or recording.

The device 100 can be delivered through a vessel adjacent to a peripheral nerve (such as shown in FIG. 35) for stimulation or recording.

The device is delivered through a vessel adjacent to a sympathetic or parasympathetic nerve for stimulation or ablation

As shown in FIG. 36, one example of a peripheral nerve (the median nerve in this example) showing possible implant location to enable neural stimulation or measurement.

FIG. 38A illustrates another example of a stent or scaffolding 101 having a plurality of electrodes 131 disposed about the stent 101 body. For purposes of illustration, the stent 101 is shown without any connecting structure that electrically couples the electrodes to leads or other such structure that allows electrical communication between the electrodes and control unit as described above. In the illustrated variation, the electrodes 131 are dispersed about the body of the stent 101 and are located at the joining or apex of joining struts 108. In such a configuration, where instead of having cells shaped like diamonds, the cells are shaped like a 'V'. This configuration can enhance the apposition between the electrodes 131 and the tissue or vessel wall.

FIG. 38A also illustrates a variation of a stent 101 that can be fabricated where stent structure comprises an integrated conductive layer that extends through a portion or more of the stent strut 108 and where the electrode 131 is formed through an exposed portion of the integrated conductive layer. Such a stent configuration, as described in detail below, permits a stent 101 electrode 131 assembly, which embeds electrodes and conductive electrode tracks into the stent lattice or strut itself. Such a construction reduces or eliminates the requirement to use fixation methods (i.e., adhesives, glues, fasteners, welds, etc.) to mount electrodes to the body of the stent. Such a construction further reduces or eliminates the need to further weld or electrically connect electrodes to wires. Another benefit is that conventional wire-connected-electrodes require accommodation of the wires about the stent struts and through the body of the stent.

FIG. 38B illustrates a stent structure 101 with integrated electrodes 131, where the stent structure is coupled to a shaft 121 at a distal end 146. The shaft, as described herein, can electrically couple the electrodes 131 to one or more control units (not shown) as described herein. In one example, the shaft 121 can comprise a guidewire, push wire other tubular structure that contains wires or conductive members extend-

ing therein and are coupled to the conductive layer of the stent at the distal end **146**. Alternatively, FIGS. **38C** and **38D** shows a variation of stents **101** that can be fabricated such that the shaft **121** is part of or integral with the stent structure, where the conductive layer extends through a portion or all of the stent to the shaft **121**. Such a construction further eliminates the need for joining the shaft to the stent structure at the working end of the stent. Instead, the joining of the stent structure (forming the shaft) to a discrete shaft can be moved proximally along the device. Such a construction allows the working end of the stent and shaft to remain flexible. The stent structures shown in FIGS. **38C** and **38D** can also include an optional reinforced section **62** as discussed above. FIG. **38C** further illustrates a hollow shaft **121**, which allows insertion of a stylet **123** there-through to assist in positioning of the device or permits coupling of wires or other conductive members there-through. Furthermore, the shaft **121** can include any number of features **119** that improve flexibility or pushability of the shaft through the vasculature.

The electrical connection of the electrodes **131** to leads extending through the device can be accomplished by the construction of one or more connection pads (similar in construction to the electrodes described below) where the size of the pads ensures sufficient contact with the wire/lead, the type of pads ensures robustness and reduces track fatigue when crimped and attached. The section containing the pads can be compressed into a tube at, for example, distal section **146** to enable insertion of a cable **121**.

In certain variations, the connection pads should be able to feed through the catheter. Furthermore, the connection pads **132** can include one or more holes or openings that enable visual confirmation that the pads are aligned with contacts on the lead. These holes/openings also enables direct/laser welding or adhesion of the contact leads (inside tube **121**) and the contact pads (on the inside of the tube spanning through the hole to the outside)

In one example, a coaxial-octofilar cable (i.e. an inner cable with 8 wires positioned inside an outer cable having 8 wires) is used to enhance fatigue resistance and to ensure that wires can fit within constraints (i.e., can be inserted through a sufficiently small catheter, and can have an internal stylet as required).

FIGS. **39A-49C** illustrate one example of a stent structure **101** constructed with an embedded electrode and conductive path. FIG. **39A** illustrates an example of a stent structure **101** in a planar configuration with electrodes **138** in a linear arrangement for purposes of illustration only. Clearly, any configuration of electrodes is within the scope of this disclosure. Specifically, in those variations of stent structures useful for neurological applications, the stent structure can comprise a diameter that is traditionally greater than existing neurological stents. Such increased diameter can be useful due to the stent structure being permanently implanted and while requiring apposition of electrodes against the vessel/tissue wall. Moreover, in some variations, the length of such stent structures can include lengths up to and greater than 20 mm to accommodate desired placement along the human motor cortex. For example, variations of the device require a stent structure that is sufficiently long enough to cover the motor cortex and peripheral cortical areas. Such lengths are not typically required for existing interventional devices aimed at restoring flow or addressing aneurysms or other medical conditions. In addition, in certain variations, the electrical path between certain electrodes can be isolated. In such a case, the electrically conductive material **50** can be omitted from certain stent

struts to form a pattern that allows an electrode to have an electrical conduction path to a contact pad or other conductive element but the electrical conduction path is electrically isolated from a second electrode having its own second electrically conductive path.

Placement of the electrodes in a specific pattern (e.g., a corkscrew configuration or a configuration of three linear (or corkscrew oriented) lines that are oriented 120 degrees from each other) can ensure a deployed electrode orientation that directs electrodes towards the brain. Once implanted, orientation is not possible surgically (i.e., the device will be implanted and will be difficult if not impossible to rotate). Therefore, variations of the device will be desirable to have an electrode pattern that will face towards the desired regions of the brain upon delivery.

Electrode sizing should be of a sufficient size to ensure high quality recordings and give large enough charge injection limits (the amount of current that can be passed through the electrodes during stimulation without damaging the electrodes which in turn may damage tissue). The size should also be sufficient to allow delivery via a catheter system.

FIGS. **39B** and **39C** illustrates a cross-sectional view of the stent structure of FIG. **39A** taken along line **39B-39B** to further illustrate one variation of a manufacturing technique of using MEMS (microelectrical mechanical systems) technology to deposit and structure thin film devices to fabricate a stent structure with electrodes and a conductive path embedded into the stent lattice or struts. The spacing of the struts in FIGS. **39B** and **39C** are compressed for illustrative purposes only.

As discussed above, embedding the electrode and conductive path presents advantages in the mechanical performance of the device. Furthermore, embedding of electrodes provides the ability to increase the number of electrodes mounted on the structure give that the conductive paths (30-50 μm ×200-500 nm) can be smaller than traditional electrode wires (50-100 μm).

Manufacture of thin-film stents can be performed by depositing Nitinol or other superelastic and shape memory materials (or other materials for deposition of electrodes and contacts (including but not limited to gold, platinum, iridium oxide) through magnetron sputtering in a specific pattern (**56**) using a sacrificial layer (**58**) as a preliminary support structure. Removal of the support structure (**54**) enables the thin film to be further structured using UV-lithography and structures can be designed with thicknesses corresponding with radial force required to secure the electrodes against a vessel wall.

Electrical insulation of electrodes is achieved by RF sputtering and deposition of a non-conductive layer (**52**) (eg, SiO) onto the thin-film structure (**54**). Electrodes and electrode tracks (**50**) are sputter deposited onto the non-conductive layer (using conductive and biomedically acceptable materials including gold, Pt, Ti, NiTi, PtIr), with an additional non-conductive layer deposited over the conductive track tfor further electrical isolation and insulation. As shown, conducting path **50** is left exposed to form the electrode **138** (similarly, a contact pad area can remain exposed). Finally, the sacrificial layer **56** and substrate are removed leaving the stent structure **101** as shown in FIG. **39C**.

In certain variations where the base structure **54** comprises superelastic and shape-memory materials (i.e. Nitinol), the stent structure **101** can be annealed in a high vacuum chamber to avoid oxidation during the annealing process. During heat treatment, the amorphous Nitinol struc-

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ture **54** crystallizes to obtain superelasticity and can be simultaneously shape set into a cylindrical or other shape as desired. The structure **101** can then be heat treated.

FIG. **40A**, which is a partial sectional view of taken along lines **40A-40A** of FIG. **41B**, illustrate an additional variation of a stent structure **101** fabricated via MEMS technology where one or more stent struts **108** can be dimensionally altered to impart desired structural or other aspects to the stent structure **101**. For example, in the illustrated variation, certain stent struts **108** are dimensionally altered such that the support material **60** comprises a greater thickness than adjacent stent structures **108**. However, such dimensional variation is not limited to thickness but can also include width, shape, etc.

FIG. **40B** illustrates the stent structure **101** resulting from the dimensionally altered struts resulting in a sinusoidal section **62** of the stent structure **101** that comprises a greater stiffness (resulting from the increased thickness). Such a configuration allowing the stent device to be pushed through a catheter rather than conventional requirements to be unsheathed (where the sheath is pulled back over the stent). Conventional stents are made from a thin lattice of Nitinol diamonds or cells. This sinusoidal section **62** can function like a backbone and gives forward pushing strength to the device without restricting super-elasticity and the ability for the stent to compress and expand. Clearly, any number of variations of dimensionally altered strut sections are within the scope of this disclosure.

Many modifications will be apparent to those skilled in the art without departing from the scope of the present invention.

The reference to any prior art in this specification is not, and should not be taken as, an acknowledgment or any form of suggestion that the prior art forms part of the common general knowledge in Australia

In this specification and the claims that follow, unless stated otherwise, the word “comprise” and its variations, such as “comprises” and “comprising”, imply the inclusion of a stated integer, step, or group of integers or steps, but not the exclusion of any other integer or step or group of integers or steps.

References in this specification to any prior publication, information derived from any said prior publication, or any known matter are not and should not be taken as an acknowledgement, admission or suggestion that said prior publication, or any information derived from this prior publication or known matter forms part of the common general knowledge in the field of endeavour to which the specification relates.

We claim:

1. A medical device for delivery from a tubular body, the medical device comprising:

a frame structure moveable between a reduced profile and an expanded profile in which a diameter of the frame structure increases;

a plurality of struts formed by the frame structure, wherein the plurality of struts comprises an electrically conductive material that extends along the strut and being covered with a non-conductive material; and

wherein one or more portions of the electrically conductive material is exposed at a first portion of an external surface of one or more of the plurality of struts to form at least one electrode, wherein the non-conductive material covers a second portion of the external surface of one or more of the plurality of struts, wherein the second portion and the first portion are spaced apart radially at a cross section of the at least one electrode.

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2. The medical device of claim **1**, further comprising a lead extending away from the frame structure, wherein the electrically conductive material extends along the strut to electrically couple with the lead, and wherein a proximal end of the lead is electrically coupleable to an implantable control unit.

3. The medical device of claim **2**, further comprising a connector block configured to electrically couple the medical device to an external device, where the lead extends from the frame structure to the connector block.

4. The medical device of claim **1**, wherein the at least one electrode has an electrode surface configured to conform to a shape of a vessel wall within the tubular body.

5. The medical device of claim **1**, further comprising a stent shaft coupled to the frame structure, wherein the stent shaft is configured to assist positioning of the frame structure within the tubular body.

6. The medical device of claim **1**, wherein the non-conductive material partially embeds the electrically conductive material.

7. The medical device of claim **6**, where the non-conductive material of at least one of the plurality of struts completely covers the electrically conductive material.

8. The medical device of claim **6**, where the frame structure further comprises a support material in contact with the non-conductive material.

9. The medical device of claim **8**, where non-conductive material is tubular shaped or c-shaped at a cross-section of the strut.

10. The medical device of claim **1**, wherein the at least one electrode comprises a plurality of electrodes.

11. The medical device of claim **10**, where the plurality of electrodes are aligned in a linear pattern on the frame structure selected from the group consisting of a linear pattern, a sinusoidal pattern, and a circumferential pattern.

12. The medical device of claim **1**, wherein the at least one electrode comprises a greater surface area than an adjacent region of the strut.

13. The medical device of claim **1**, where the at least one strut comprises a first strut and further comprising a second strut forming the frame structure, where the second strut comprises a conductive material and a non-conductive material wrapping partially around the conductive material, where the conductive material of the first strut is electrically isolated from the conductive material of the second strut; and

at least a second electrode formed by an opening in the non-conductive material on the second strut.

14. The medical device of claim **1**, further comprising at least one reinforced strut having a support material having a width or thickness greater than at least a second strut, the reinforced strut extending along a length of the frame structure.

15. The medical device of claim **1**, further comprising at least one reinforced strut having a support material having a width or thickness greater than at least a second strut.

16. The medical device of claim **15**, where the reinforced strut extends along a length of the frame structure.

17. The medical device of claim **1**, wherein a cross link between adjoining struts terminates in an apex, where the electrode is located on the apex to enhance apposition between the electrode and the tubular body.

18. The medical device of claim **1**, further comprising a control unit configured to generate a signal that controls an external apparatus, wherein electrical activity transmits from the plurality of electrodes to the control unit.

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19. The medical device of claim **18**, wherein the control unit is configured to wirelessly communicate with the external apparatus.

20. The medical device of claim **18**, wherein the external apparatus comprises one or more of an exoskeleton, a prosthetic limb, a wheelchair, a computer, and an electrical or electro-mechanical device.

21. A medical device for delivery from a tubular body, the medical device comprising:

a structure;

at least one strut formed by the structure, wherein the at least one strut comprises an integrated conductive layer extending throughout the structure; and

wherein the at least one strut comprises at least one electrode formed by exposing a first portion the integrated conductive layer, wherein the at least one electrode is not affected during movement of the structure when implanted, wherein the first portion is spaced apart radially from a second portion at a cross section

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of the at least one electrode, wherein the second portion is covered by a non-conductive material.

22. A medical device for delivery from a tubular body, the medical device comprising:

a frame structure moveable between a reduced profile and an expanded profile in which a diameter of the frame structure increases;

a plurality of struts formed by the frame structure, wherein the plurality of struts comprises an integrated conductive layer extending throughout the frame structure; and

wherein at least one of the plurality of struts comprises at least one electrode formed by exposing the integrated conductive layer, wherein the at least one electrode is located at a cross link between joining struts such that an integrity of the at least one electrode is not affected during movement of the frame structure between the reduced profile and the expanded profile.

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